

User guide

RectChr

RectChr: Interactive multi-track visualization of genomic statistics on rectangular chromosome layouts

chr1

chr2

chr3

A Circos-like feature that changes circles to rectangles (Rect).

The GUI version is a standalone manual; you need to refer to [another GUI manual](#)

Version 1.50 User Manual

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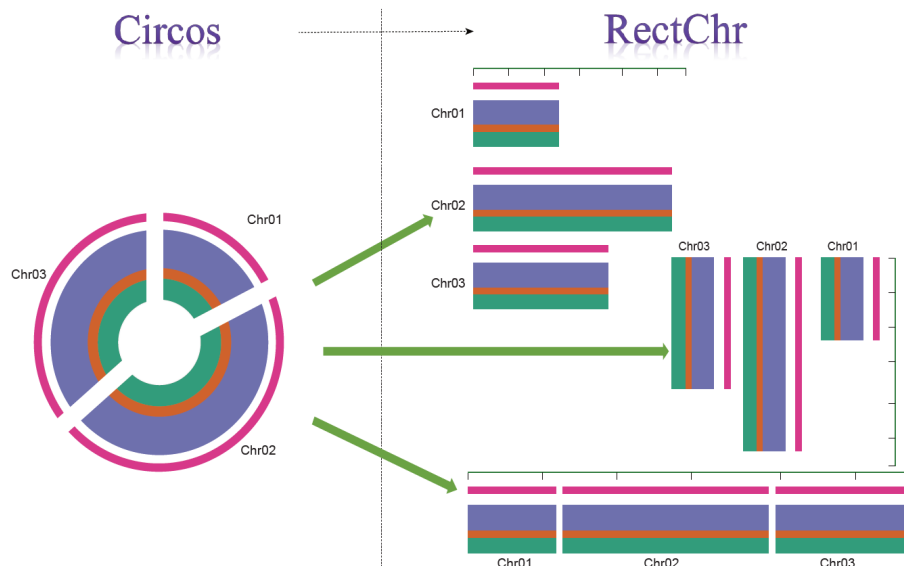
1. Introduction

[RectChr](#) is a multi-level visualization tool focused on chromosome (Chr) levels. It utilizes a range of **visual elements** and **color mappings** to display statistical variables across chromosomal regions. Supported representations include **scatter plots (scatter/points)**, **shapes (shape)**, **lines (line)**, **histograms (histogram)**, **heatmaps**, **highlighting**, **text**, **ridgelines**, **rainbow links**, **self-linking (LinkS)**, **animated heatmaps**, **animated histograms**. These views help users identify patterns in their data and interpret analysis results.

A major advantage of RectChr is its high flexibility. Users can combine visual elements and customize their parameters. Its workflow is similar to that of the classic Circos tool, but it presents the visualization in a rectangular layout, with chromosomes arranged horizontally or vertically. Visualizations that can be produced with Circos can also be created with RectChr in this layout. Colors and styles can be configured independently for each track.

RectChr also provides default settings that simplify common tasks. For analyses such as SNP or GC density, users can supply the input file directly to generate a visualization with minimal configuration.

An **interactive GUI** is available in the latest version, **1.50**. See the GUI user manual.



The package includes 10 application examples. A **Zhihu** column also collects user-contributed configurations and explanations.

New features by version

- A. 1.29 Added ColorBrewer color palettes.
- B. 1.30 Fixed a bug introduced in 1.29 that required manual TopVHigh/TopVLow se for certain datasets to avoid an error.
- C. 1.31 Added the shape plot type, supporting 12 shapes.
- D. 1.33 Added the ridgeline plot type. Click to view the feature description.
- E. 1.35 Added `zoom_region` to zoom in on a selected region. Click here for details.
- F. 1.39 a: Introduced new parameter names and `ParaOld2New.pl` to convert old configuration files to the new format.
 - b: Added left and right semicircles to the shape plot type.
 - c: Added `track_shift_y` to overlay multiple plot types within the same apparent track.
- G. 1.41 legend adds a total of 13 layout methods
- F. 1.50 Added a GUI with Windows support.

2. Function Overview

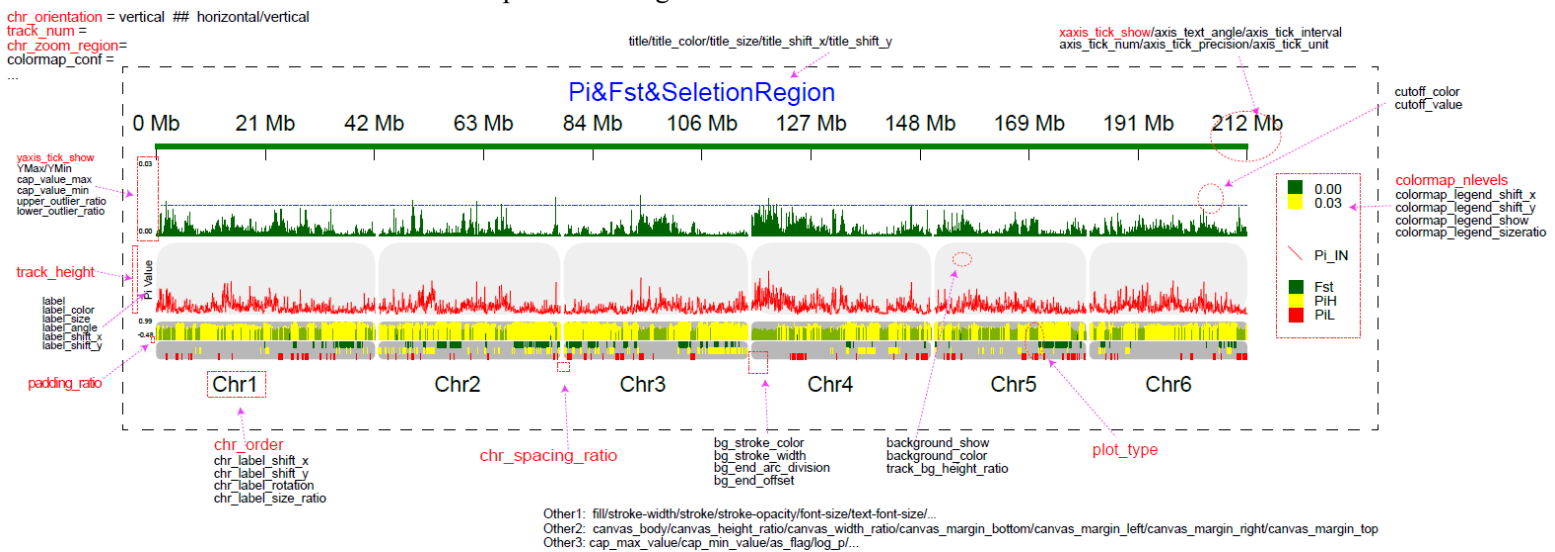
RectChr provides the following flexible visualization features:

- 1) Custom chromosome layout:** Set the chromosome orientation (*chr_orientation*) to horizontal or vertical, and specify the chromosome order (*chr_order*). Customize spacing (*padding_ratio*), track height (*track_height*), background color (*background_color*), and other layout settings.
- 2) Multiple tracks:** Define multiple tracks for each chromosome using *track_num*(level), with a different plot type for each track.
- 3) Plot types:** Choose from 12 plot types (*plot_type*): scatter/point, shape, line, histogram, heatmap, highlights, text, ridgeline, PairWiseLink, LinkS, heatmapAnimated, and histAnimated.
- 4) Colors and data ranges:** Customize the color palette (for example, *colormap_brewer_name*), color gradients, and number of color levels (*colormap_nlevels*). Limit the data range using parameters such as *YMax*, *upper_outlier_ratio*, and *cap_max_value*.
- 5) Consistent input format:** The first three columns identify the genomic region. Specify the statistics to plot with a simple column reference, such as *show_columns* = File2:4, which uses the fourth column of the second file.
- 6) Regional zoom:** Use *zoom_region* to focus on a specific region and inspect details.
- 7) Parameter customization:** All parameters are available for users to adjust the visualization to

their needs.

8) ...

Below are some control parameter diagrams:



Some important parameters are highlighted in red.

3. Plot Type Examples

This section introduces basic usage. The corresponding data and configurations are provided in the subdirectories of **Basic_Tutorials** in the software package. The following sections introduce the 12 plot types.

No.	Plot type	plot_type	Description	Example (Basic_Tutorials)	Location
1	Heatmap	heatmap	Color intensity represents data density or magnitude, revealing distributions and trends. Common uses include gene expression and SNP density.		example01
2	Histogram	histogram	Bar height and color represent numerical values. Commonly used to compare groups, such as nucleotide diversity (pi) in two populations.		example02
3	Points	scatter/point	Point height and color represent values. Each point is a data item. Useful for distributions, outliers, and associations, such as GWAS.		example03
4	Lines	line	Connect data points to show continuous trends along chromosomes, such as Fst.		example04
5	Highlights	highlights	Highlight selected genomic regions or elements with color to draw attention to features such as centromeres and QTL regions.		example05
6	Text	text	Add text annotations, optionally colored, to label data points, explain the plot, or provide additional information.		example06
7	Shapes	shape	Represent data using 15 shapes, including circles, squares, and triangles. Shape and color can encode different data features.		example07
8	Ridgeline	ridgeline	Stack multiple distribution curves to compare their shapes, central positions, and dispersion.		example08
9	Pairwise links	pairwiselink pairwiselinkV2	Show relationships between data objects at two points on a chromosome. Curved links connect related objects; color and height can encode association strength or other attributes.		example09
10	Self-links	LinkS	Show relationships between two regions or data objects on a chromosome. Colored links connect related objects; color and height can encode association strength or other attributes.		example10
11	Animated heatmap	heatmapAnimated	Animate heatmap colors over time in an SVG to show data distributions and changes across time points or states. Less commonly used.	Open the example SVG in a browser.	example11
12	Animated histogram	histAnimated	Animate bar heights or colors over time in an SVG to show changes across time points or states. Less commonly used.	Open the example SVG in a browser.	example12
13	Other	OthersNA	An unrecognized plot_type draws only the background bar, which can serve as a placeholder or divider.		-

To illustrate the differences between plot types, most examples in **Basic_Tutorials** use a single track (*track_num*=1), analogous to one ring in Circos. See the example directories under **Basic_Tutorials** for the data and configuration files.

3.1 Heatmap Examples

Example 1

Input format: [#Chr Start End Flag] , with four columns, Flag contains the statistic for the region and may be an integer, a decimal value, or a string. In the plots below, colored bins represent the values.

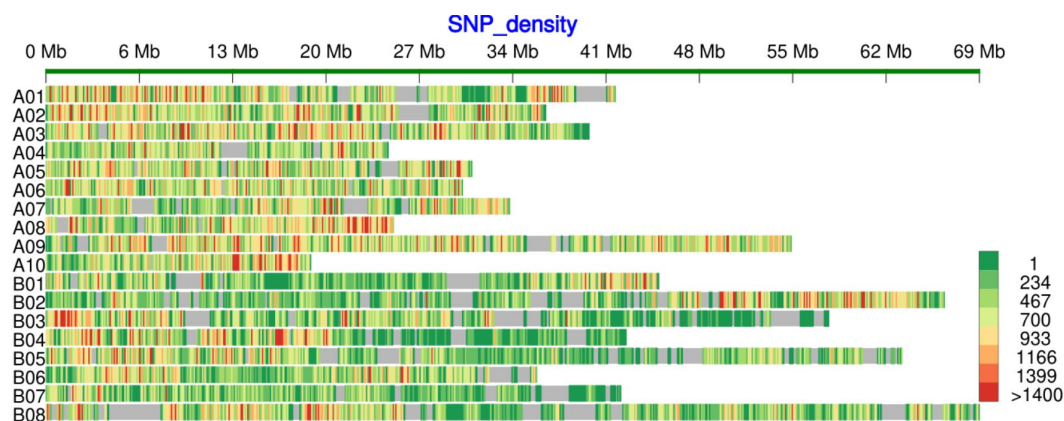
Density plot:

- 1 Chromosomes are arranged vertically by default: *chr_orientation*=vertical.
- 2 The number of tracks, analogous to the number of Circos rings, is *track_num*=1 here. By default, it is inferred from the number of input columns.
- 3 The default *plot_type* is heatmap.
- 4 The default data source is *show_columns*=File1:4, using the fourth column of File1.

All these settings can therefore be left at their defaults; only the input file is required.

```
SetParaFor= global
File1= ./SNP.dis.gz
```

See **Basic_Tutorials/example01** for the configuration and data.



The **example01_Density_heatmap** directory includes a Perl script, **CountBinNumPlot.pl**, that generates the SNP density plot above directly from a VCF file. It can also generate a gene density plot directly from a GFF file. See application example01 below.

```
#### run SNP density for VCF File example ####
#perl ../bin/script/CountBinNumPlot.pl -InFile in.vcf.gz -OutPut SNP_density
#perl ../bin/script/CountBinNumPlot.pl -InFile in.vcf.gz -OutPut SNP_density -BinSize 100000
#### run gene density for gFF File example ####
#cat Ref.gff|awk '3=="mRNA">mRNA.gff'
#perl ../bin/script/CountBinNumPlot.pl -InFile mRNA.gff -SiteColumn 4 -OutPut gene_density
```

Example 2

If *track_num* is not specified, it is inferred from the number of columns in File1.

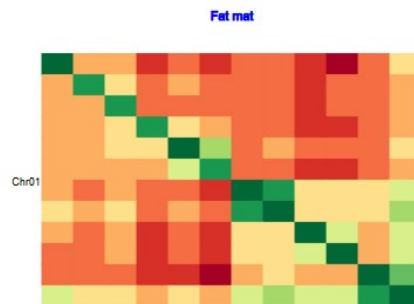
For example, mat.fst contains chr, start, end, and Flag1 through Flag12. The default is therefore *track_num*=12, with each trackX displaying FlagX. Use the settings below to produce a

heatmap with a shared *Ymax* and *Ymin*.

```
SetParaFor= global
File1= mat.fst
title= "Fat mat"
#track_num    =12          ## 默认为根据 File1 的列数减 3
canvas_body=360           ## 12 列*20    mat.fst

xaxis_tick_show=0         ##不显示 x 轴
SetParaFor= trackALL
plot_type= heatmap        ##作图方式用 热图,文本
upper_outlier_ratio=1.0
colormap_nlevels=10
Ymax = 0.65 ##设置此层(ValueX) maximum value, default automatic, displayed range
Ymin=0 ##设置此层(ValueX) minimum value, default automatic, displayed range
colormap_legend_show=0
```

The results are as follows:



3.2 Histogram Examples

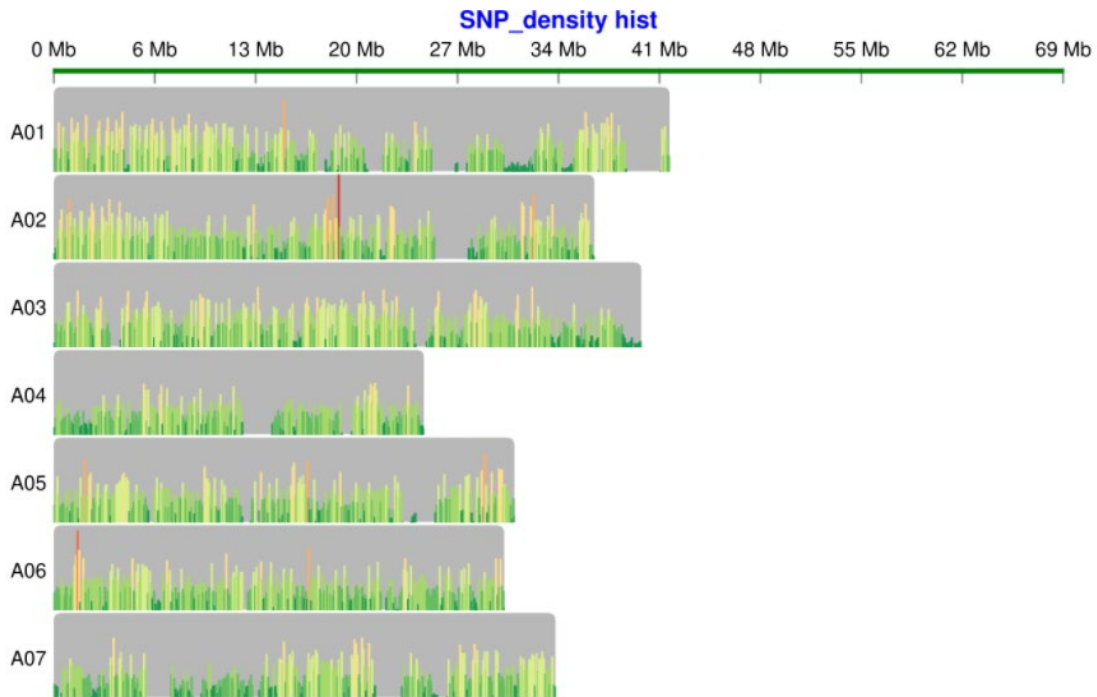
Example 1

Use the same data as in **example01_Density_heatmap**.

Density plot:

- 1 Chromosomes are arranged vertically by default.
- 2 The number of tracks, analogous to the number of Circos rings, is *track_num*=1 here. By default, it is inferred from the number of input columns.
- 3 Change *plot_type* to hist.
- 4 Increase the current track height: *track_height*=100.

See **Basic_Tutorials/example02** for the configuration and data.



Example 2

Further customize the plot as follows.

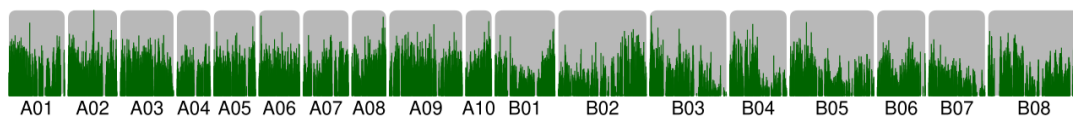
1. Change the chromosome arrangement from vertical to horizontal: `chr_orientation=horizontal`.
2. To use a single color instead of a gradient:
3. Set the number of color levels: `colormap_nlevels=3`.
4. Set all three gradient colors to the same value.

```
colormap_low_color = "#FFFF00" ## The color corresponding to the lowest value
colormap_mid_color = "#FFFF00" ## The color corresponding to the middle value
colormap_high_color = "#FF0000"
colormap_legend_show=0 ##不显示渐变条,不显示图例
```

5. Hide the X axis.

```
axis_tick_show=0 # no display xaxis
```

This produces the following plot:

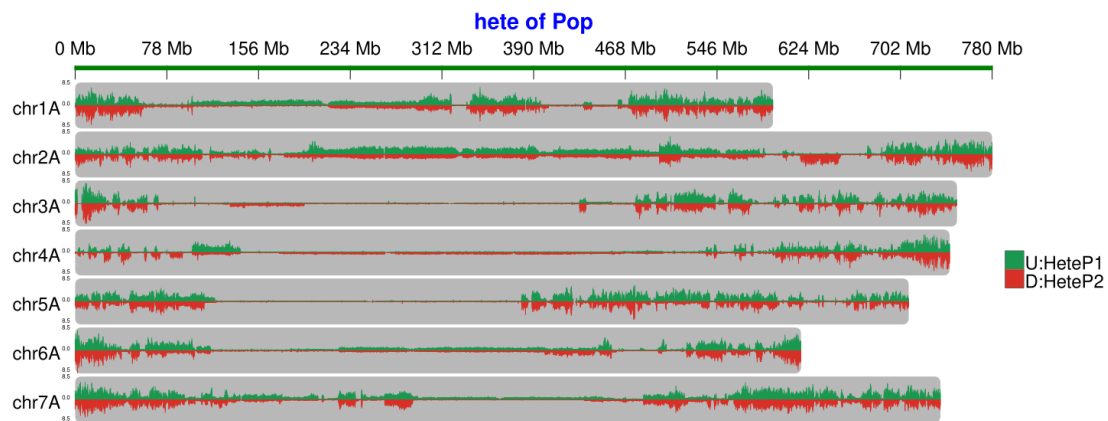


Example 3

For comparison, two data columns can be displayed as opposing histograms in the same track, one extending upward and the other downward from the center.

Data source: `show_columns=File1:6,7` uses columns 6 and 7 of File1 for the upper and lower histograms.

```
SetParaFor = track1 ## Start processing the first-layer parameters
plot_type = hist ## plot type: histogram(直方图)
show_columns = File1:6,7 ## 用 File2 的第 6,7 列数画上下直方图
yaxis_tick_show=1 ## 显示 y 轴数值
track_height=60 ## track 高度高些
#bg_end_arc_division=8 ## 背景边缘曲度,值越大曲度小些
```



3.3 Scatter Plot Examples

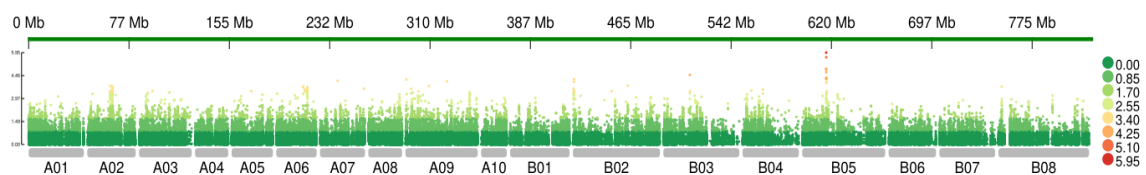
Example 1

This GWAS scatter plot uses two tracks: a taller data track and a shorter background track. By default, point color and height are determined by the values. Configure it as follows:

```
track_num=2                                ## 共二个 track(层)
SetParaFor=track1 ## Now start processing the first layer parameter processing
plot_type=point                             ##作图方式散点
show_columns=File1:4                       ##把 file1 的第四列用散点图形式画出来)
track_height=100 ## Raise the width of the first layer
log_p=1                                    ##0-log(), 对数值进行取 log
background_color="FFFFFF" ##背景为白色, actually not drawn
yaxis_tick_show=1                          ##显示左边数值
padding_ratio=0.04 ##和 the next layer of track, leave a blank space
track_point_size=1.2                       ##散点的大小按默认放大

SetParaFor=track2 ## Draw nothing on this layer, just use the background bar
show_columns=File1:4
track_height=10
plot_type=EEE                              ##啥都不画,空,仅背景条
```

This produces the plot below:



Example 2

In the Manhattan plot above, colors depend on the data values. To color the points by chromosome instead, first note the following behavior for point, heatmap, and histogram plots:

A: With `show_column=File1:4`, colors are assigned to the values in column 4 of File1 according to the color-level rules.

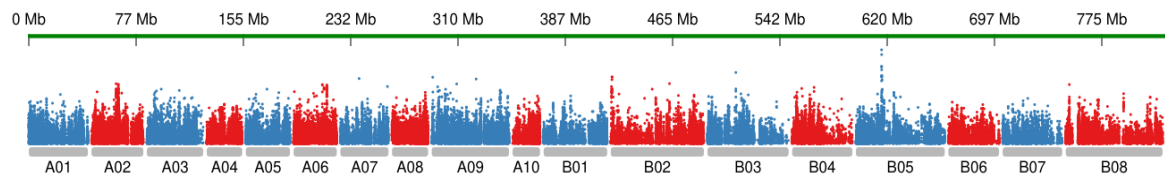
B: With `show_column=File1:4 File2:4 File2:5`, column 4 of File1 uses color A, column 4 of File2 uses color B, and column 5 of File2 uses color C.

```
zcat Gwas.data.pvalue.gz | grep "A02|A04|A06|A08|A10|B02|B04|B06|B08" > A.pvalue
zcat Gwas.data.pvalue.gz | grep -v "A02|A04|A06|A08|A10|B02|B04|B06|B08" > B.pvalue
```

The main change is therefore:

```
ShowColumn = File2:4 File3:4      ## File2:4 列颜色 A      File3:4 颜色 B
```

The final result is shown below:



Large GWAS datasets can produce large SVG files and require substantial peak memory when converting to PNG. Adjacent SNPs may be in complete linkage disequilibrium (LD) and have identical P values (Y coordinates). Their X coordinates may also be indistinguishable at the plotting scale: for a 500 Mb genome on a canvas of 5,000 units, one canvas unit represents 100 kb, so 1,000 bp corresponds to only 0.01 canvas units. Such points are rendered at effectively the same position. From version 1.24 onward, RectChr retains only the last point at an overlapping position, reducing SVG file size without changing the visible plot.

3.4 Lines Plot Examples

Example 1

```
SetParaFor= global
File1= ./poly.info.gz      ##这个是必须输入文件路径
track_num = 1 ##多少层, how many circles does the circos have?
SetParaFor=track1          ##下面开始处理第 2 层参数处理
plot_type= lines           ##作图方式用线条
show_columns= File1:4,5    ##把 file1 的第四和五列用 lines 图形式画出来
```



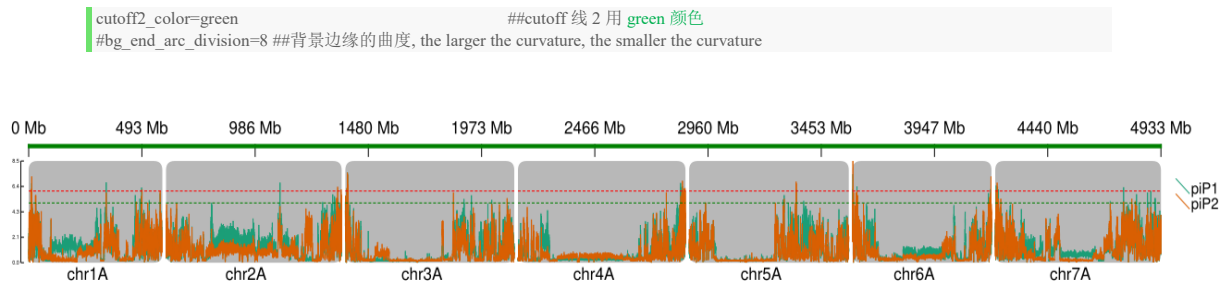
Example 2

Cutoff lines can be added to point, histogram, and line plots. The following example includes cutoff lines.

Horizontal layout with two cutoff lines:

```
SetParaFor= global
File1= ./poly.info.gz      ##这个是必须输入参数, 输入文件路径
track_num = 1 ##多少层, how many circles does the circos have?
chr_orientation= horizontal ##染色体排列方向为水平(horizontal),横排
#canvas_height_ratio=0.8   ##画布下面空白去掉些

SetParaFor=track1          ##下面开始处理第 2 层 参数处理
plot_type= lines           ##作图方式用线条
show_columns = File1: Columns 4,5 ##把 file1 的第 4,5 列 are drawn in line diagram format).
track_height=100
yaxis_tick_show = 1 ##显示的 Y axis start and end values
cutoff_y=6.0
cutoff2_y=5.0 ##cutoff_y, cutoff1_y, cutoff2_y can draw three lines
```



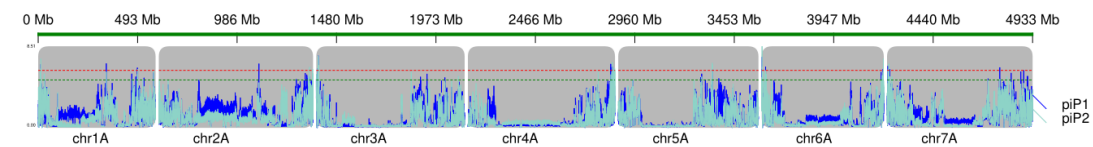
Example 3

When plotting several lines, you can emphasize an individual line.

Use `line_colors_conf=line_colors.conf` to specify a color for each line. The contents of `line_colors.conf` are shown below.

1=blue

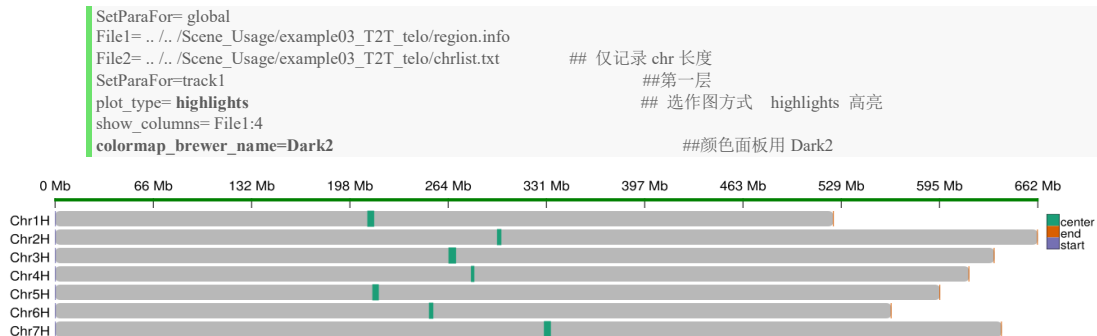
2="#1B9E77"



3.5 Highlights Examples

Example 1

As shown below:



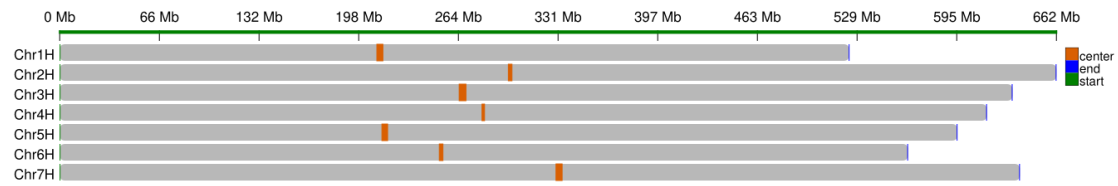
Example 2

To assign colors manually, define the relationship between each value (Flag) and its color in the global settings.

`colormap_conf= col.file` This parameter works in all drawing methods. The contents of `col.file` are as follows:

```
center="#D95F02"
end=blue
start=green
```

Thus, the following results can be obtained:



3.6 Text Example

Example 1

Input format: #Chr Start End ScaName +/-

This example combines two tracks: text and highlights.

```
SetParaFor= global
File1= ./scaf2chr.format2          ##File1 是必须输入参数, 输入文件路径
track_num= 2 ##设 2 层,类同 circos 多少个圈,
chr_spacing_ratio=0.2              ##不同染色体 chr 之间的间隔比例(ChrWidth*ChrSpacingRatio)
title= "Scaf2Chr" ##标图 text, along with other control parameters

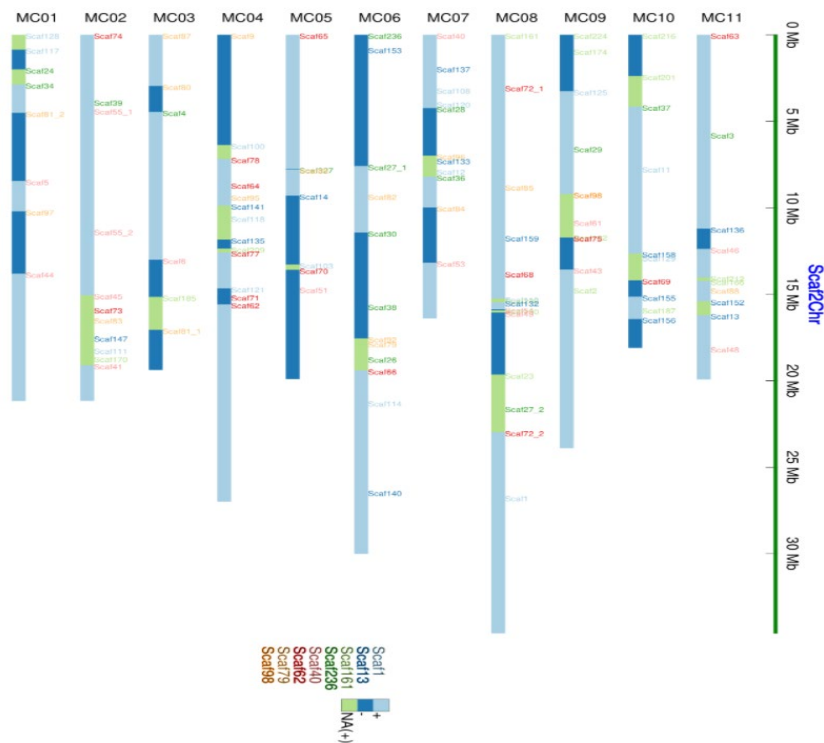
canvas_angle= 90                   ## 对 Figure png 进行旋转的角度
chr_label_rotation= -90            ##旋转 chr 名字 text
axis_tick_interval=5000000         ##X 坐标以 5M 为单位

SetParaFor= trackALL ## 下面是处理初始化参数 SetParaFor 参数处理
#colormap_legend_shift_x=-180
#colormap_legend_shift_y=-400

SetParaFor=track1                  ##下面开始处理第 1 层 参数处理
plot_type= text
show_columns= File1:4
track_height=80 ##放 text 的名字 width length
background_color="FFFFFF" ##白色, meaning no background
#track_text_angle=-90              ##旋转 90 度文字
text-font-size=14 ##设置文字大小

SetParaFor=track2                  ##下面开始处理第 2 层 参数处理
plot_type = heatmap ##热度图 means + -
show_columns = File1: Column 7 ##把 file1 的第 7 是 plotted in scatter plot form
```

See example1 for the configuration, usage, and input format. Text can be configured to use a single color.



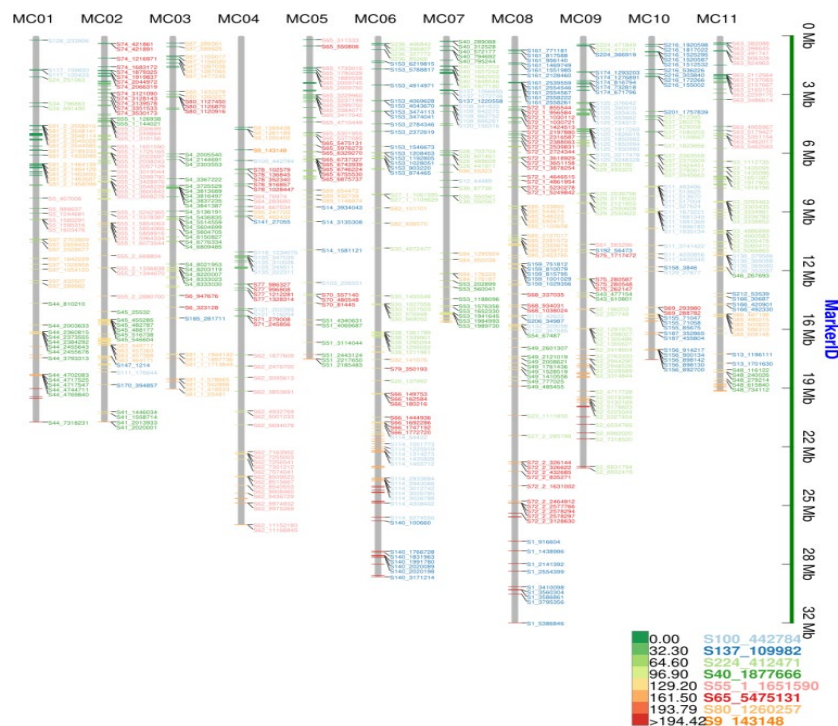
Example 2

Genetic mapping plotting

Input format: #Chr Start End ScaName +/-

Combine a text track with a highlight track. Reduce the width of the background bar in the highlight track using *track_bg_height_ratio*. Add the following setting under *SetParaFor*=track2.

```
track_bg_height_ratio=0.5    ## 背景的宽度变小点
```



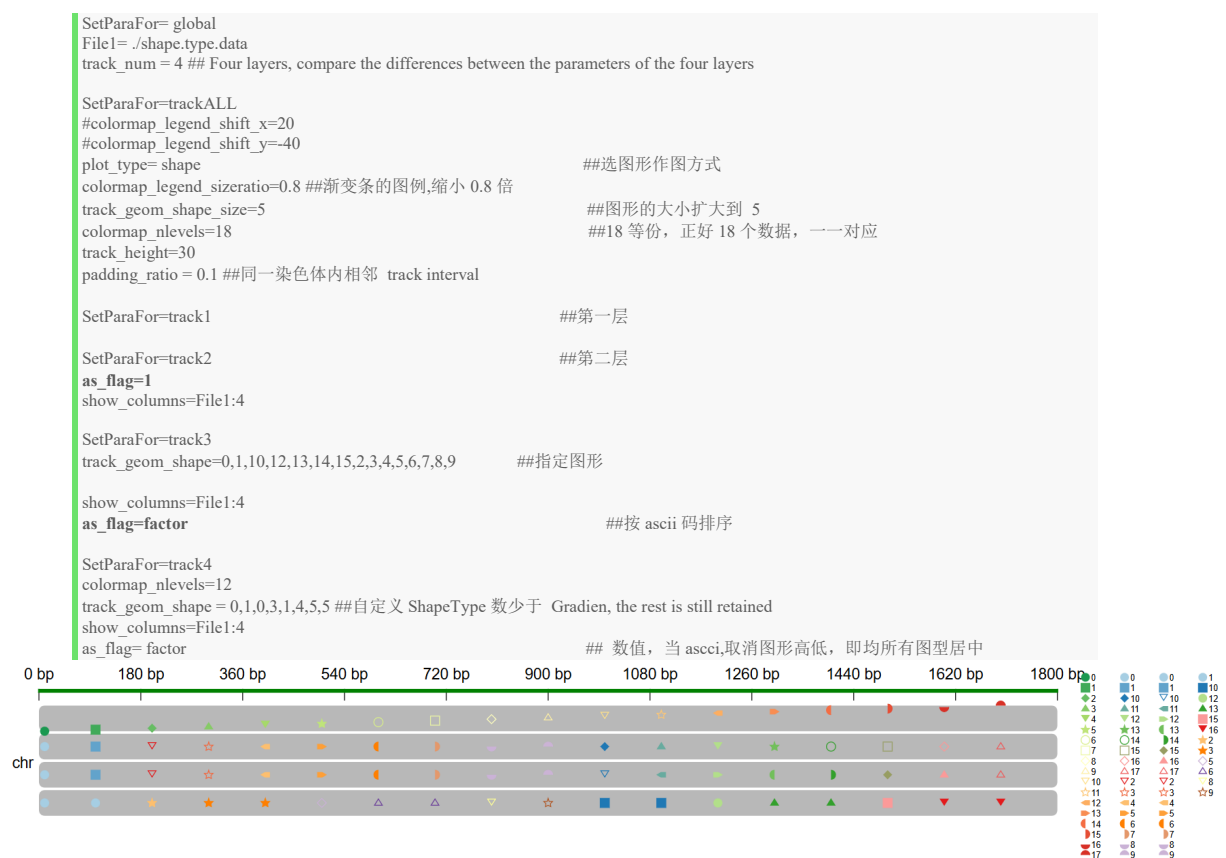
Background width, track settings, and a uniform color in place of a gradient can all be customized and combined.

3.7 Shape Examples

Example 1

Input file A format: [#Chr Stat End Flag]. As with other plot types, the program detects whether Flag contains numbers or strings and sorts the values accordingly. For numeric shape data, vertical position can also reflect the value. To treat numeric values as categories, add as_flag=factor to the corresponding track. The four tracks below all use File1:4.

The second track uses ASCII sorting. The third also uses ASCII sorting but specifies custom shapes. The shape plot type provides 16 shapes; compare the mappings in track1 and track3.



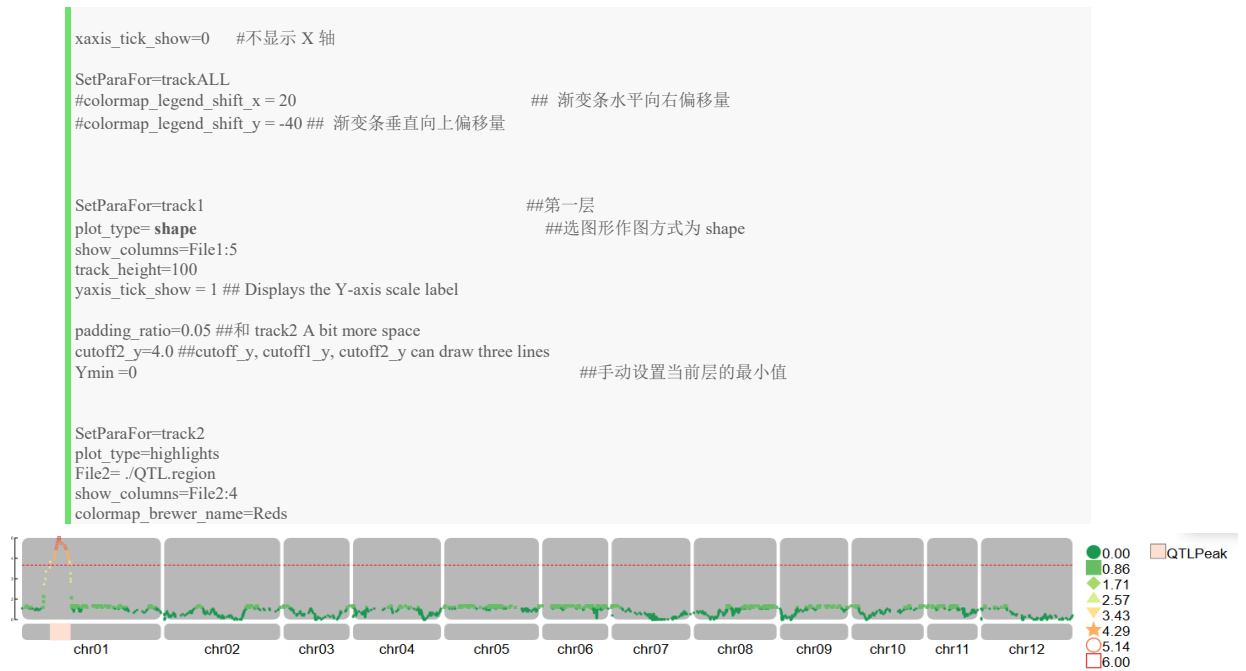
Shapes corresponding to ShapeType values 0 through 17 In order: filled circle, square, diamond, upward triangle, downward triangle, and pentagon; outline circle, square, diamond, upward triangle, downward triangle, and pentagon; left arrow, right arrow, left semicircle, right semicircle, upper semicircle, and lower semicircle.

Example 2

```

SetParaFor= global
File1= ./rod.data
track_num= 2
chr_ orientation= horizontal #
## 2 track
## horizontal chr 水平,横排列

```



3.8 Ridgeline Examples

Example 1

A ridgeline plot provides an alternative to a line plot. The example below illustrates the difference when the line representation in track1 is changed to ridgeline.

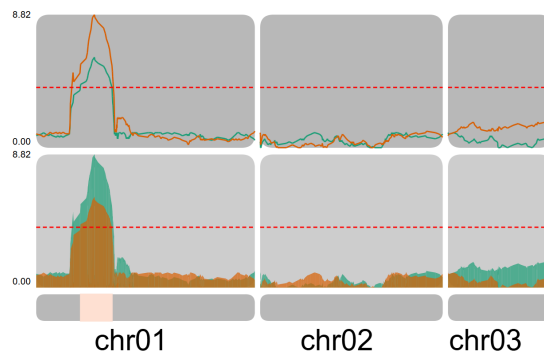
To control transparency in the SVG, set stroke-opacity and fill-opacity to 0.7 in the ridgeline track.

```

SetParaFor=track1          ## 第一层
plot_type = lines ##选图形作图方式 is line lines
show_columns=File1:5,6
track_height=100
yaxis_tick_show = 1 ##显示 Y-axis scale label
#bg_end_arc_division=10
padding_ratio=0.05 ##和 track2 A bit more space
cutoff2_y=4.0 ##cutoff_y, cutoff1_y, cutoff2_y can draw three lines
ymin = 0 ##手动设置当前层的最小值, override the automatically calculated result

SetParaFor=track2          ## 第二层
plot_type= ridgeline      ##选图形作图方式山脊线
show_columns=File1:6,5    ##第六列先画放在前面
track_height=100
yaxis_tick_show = 1 ##显示 Y-axis scale label
#bg_end_arc_division=10
padding_ratio=0.05 ##和 track2 A bit more space
cutoff_y=4.0 ##cutoff_y 画虚,默认红色
Ymin =0                ##手动设置当前层的最小值
colormap_brewer_name=Dark2
stroke-opacity=0.7      ##SVG 透明度
fill-opacity=0.7        ##SVG 透明度
stroke-width=0          ##SVG 相关属性

```



3.9 Pairwise Link Examples

Example 1

Input format: [#Chr Start End Flag] with *plot_type*=PairWiseLink.

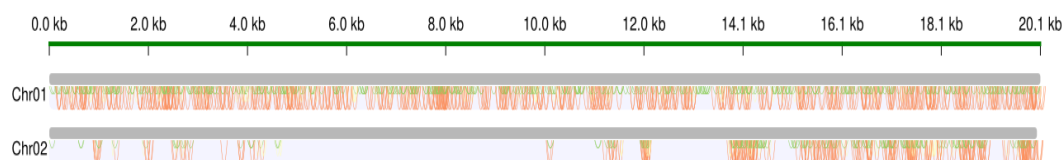
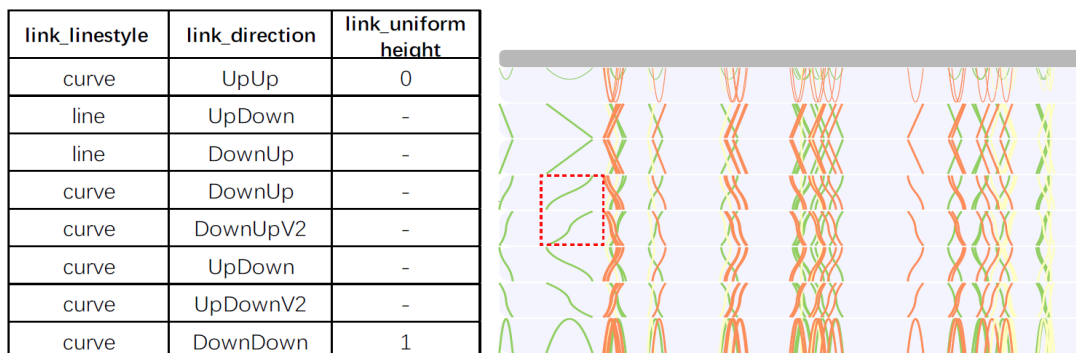
The plot below shows distances between paired-end alignments on a scaffold; see OUT1 in **example09**.

Relevant settings:

```
## link_uniform_height=1 ## link_linestyle = curve
```

For PairWiseLink, set to 1 to use a uniform height # Link style: curve (default) or line ## *link_direction* =UpUp ## For PairWiseLink, DownDown draws downward arcs; UpUp (default) draws upward arcs. UpDown and DownUp are also supported; see configuration 3 in the example and the linked explanation.

The following diagrams show the same data with different settings.

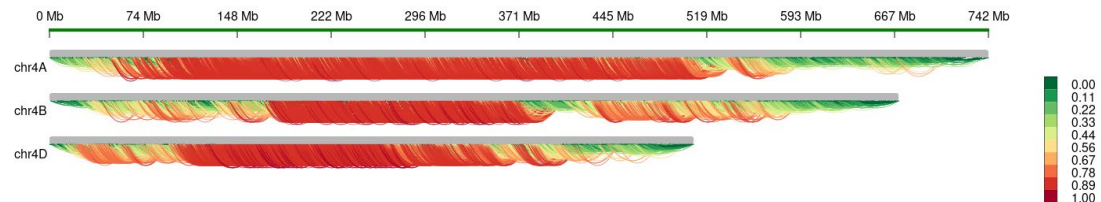


Example 2

The following R^2 plot selects one SNP per 1 Mb and displays its R^2 relationship with nearby

SNPs in the 30–40 Mb region. See OUT3 in example9.

```
SetParaFor=track2                                ##下面开始处理第 2 层 参数处理
show_columns = File1:4 ##把 file1 的第 4 列用 the color of pairwiselink
plot_type= PairWiseLink
background_color="#FFFFFF" ##背景色为#FFFFFF 为白, meaning no back color is drawn
track_height=30
stroke-width=0.5                                ##线条细点
upper_outlier_ratio=1 ## For this layer, the Top of ValueX uses the highest point color [0.95], and divides the rest equally
```



This example has two main tracks. The first draws only a background bar. The second uses PairWiseLink with a white background, making the background invisible. Its height can be customized in the configuration.

Example 3

The input format above is:

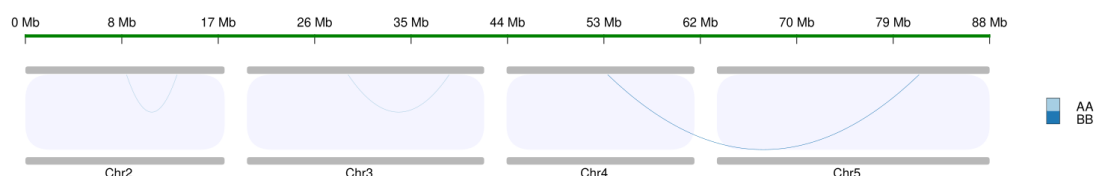
```
Chr Start End Value
```

A link runs from Start to End on the same chromosome, with Value determining the color or height. This format cannot link ChrA StartA to ChrB StartB across different chromosomes. PairWiseLinkV2 was developed to support cross-chromosome links. Its ShowColumn setting must specify three columns, such as File1:4,5,6, containing Value, ChrB, and SiteB, respectively. These columns may come from different FileX inputs, but all three are required. Other link style parameters are the same as above. For example:

```
SetParaFor=track1 ##下面开始处理第 1 layer of parameter handling, the first layer only has the back bar
track_height=10
plot_type=EEE                                ##随便一个非关 Jian Ci, indicating that it would not be painted

SetParaFor=track2                                ##第二层
plot_type= PairWiseLinkV2
background_color="#F4F4FF"                    ##背景色
track_height=100
stroke-width=0.5                                ##线条细点
show_columns=File1:4,5,6                      ##须是 3 列分别为 Value CBB SiteB
colormap_legend_show=1

SetParaFor=track3                                ##第三层
plot_type= EEE
track_height=10
```



3.10 LinkS Examples

Example 1

Pairwise links connect one position to another. LinkS extends this to links between two intervals.

The input file is:

```
ChrA StartA EndA Value chrB StartB EndB
```

ShowColumn must specify four input columns; otherwise, no links are drawn. For example, set ShowColumn as follows:

show_column=File2:4,5,6,7 or *show_column=File2:4 File2:5 File2:6 File2:7*

Use *label = "SampleA"* to set the current track label. Its style can be controlled with *label_size*, *label_color*, *label_shift_x*, *label_shift_y*, and *label_angle*.

The *chr_order* setting below selects only the listed chromosomes, omitting Chr1H and Chr4H.

```
SetParaFor = global                                ##设置当全局范围 (global)

File1 = ../Scene_Usage/example03_T2T_telo/chrlst.txt
File2 = SV.info
chr_orientation = horizontal                        ##染色体排列方向为水平 horizontal (横向)

chr_order = Chr2H,Chr3H,Chr5H,Chr6H,Chr7H ##筛选并指定染色体顺序(去掉 1H 4H)

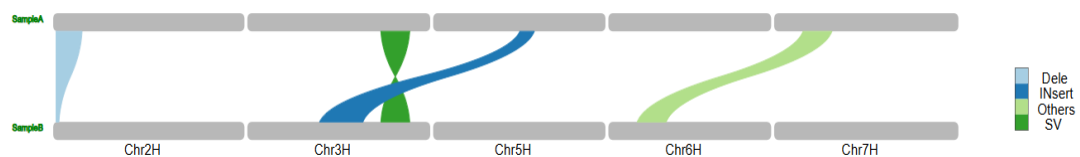
track_num = 3                                     ##指定绘图层数(track 数量)
xaxis_tick_show=0

SetParaFor = track1
plot_type = linesEE                               ##不画,仅背景条
show_columns = File1:4
label = "SampleA" ## Label of the current layer

SetParaFor = track2 ## Start processing the second-layer parameters
plot_type = LinkS
track_height = 100 ##当前层的高度 raise it a bit (i.e., orbital height).
show_columns = File2:4,5,6,7                      ##显示 linkS 须要 4 列, 均为同一文件 File2
background_show=0                                 ##不画背景

SetParaFor = track3
plot_type = linesEE                               ##不画,仅背景条
show_columns = File1: 4 ##显示 The 4th column of File1
label = "SampleB"                                 ## 当前层的标签及其样式设置 (label_size label_color ...)
```

Result:



3.11 heatmapAnimated Examples

Example 1

heatmapAnimated creates an SVG heatmap whose colors change over time or between states, displaying changes in the data distribution. Specify multiple columns with *show_columns*, as follows:

```
SetParaFor= global
File1= ./S01.bin.gz
track_num = 2 ## How many layers? How many circles is the same as circos? Set up 2 layers here
canvas_margin_right=250 ## Extend the dot on the right side of the canvas
```

```

SetParaFor= trackALL ##trackALL, which is the default value set for all layers first

SetParaFor=track1
plot_type= heatmapAnimated
show_columns= File1:4,5,6,7
padding_ratio=0.1

SetParaFor=track2
plot_type = EEE ##不画 Background bar only
show_columns= File1:4
track_height=10

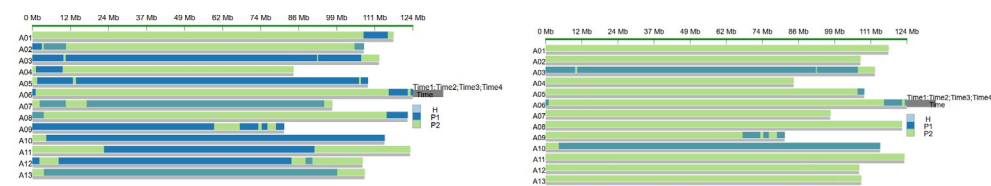
```

##选作图方式为动太热度
##必须是同一文件中的四列

##这一层 track 宽度小点

This less frequently used feature requires opening the SVG locally in a web browser to view the animation.

The four input columns represent four states. Static views of two states are shown below; see [examp11](#).



3.12 histAnimated Examples

Example 1

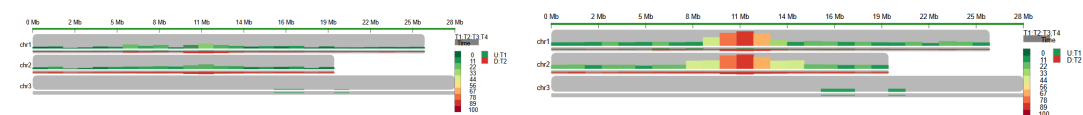
The usage is the same as heatmapAnimated: specify multiple columns with `show_columns`, but use histograms instead of heatmaps. See [example12](#). Open the exported SVG locally in a web browser to view the animation.

The four input columns represent four states. Static views of two states are shown below; see [examp12](#).

```

SetParaFor=track1
plot_type= histAnimated
show_columns= File1:4,5,6,7
padding_ratio=0.1
track_height=40
colormap_nlevels=10
upper_outlier_ratio=1.02

```



4 Application Examples

4.1 SNP and GC density plots

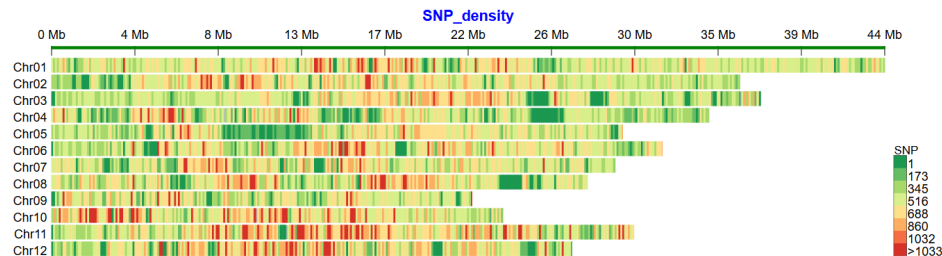
Input format: [#Chr Stat End Flag]. Flag contains the statistic for the region and may be an integer, decimal value, or string. The plots below use colored bins to represent values.

The same data can also be displayed using point heights, histograms, or lines, combined with color. The default *plot_type* is heatmap.

A minimal configuration therefore requires only the input file.

```
SetParaFor=global
File1= ./SNP.dis.gz
```

See [Basic_Tutorials/example01](#) for the configuration and data.



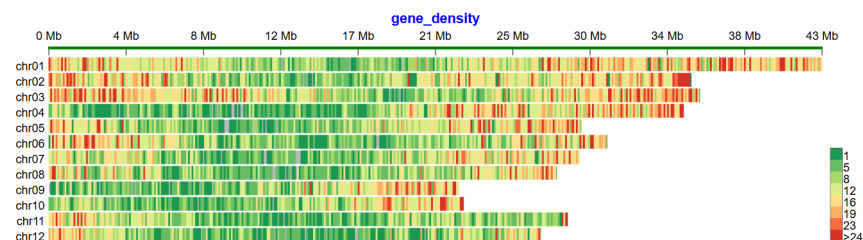
The **example01_Density_heatmap** directory provides **CountBinNumPlot.pl**, a Perl script that generates the plot above directly from a VCF file.

```
##### SNP/InDel density #####
perl CountBinNumPlot.pl -InFile in.vcf.gz -OutPut SNP_density
#perl CountBinNumPlot.pl -InFile in.vcf.gz -OutPut SNP_density -BinSize 100000
```

The script has additional options and is not limited to VCF input. It can sum or average the values of a specified column within bins.

```
##### gene density #####
cat Ref.gff |awk ' $3=="mRNA">mRNA.gff'
perl CountBinNumPlot.pl -InFile mRNA.gff -SiteColumn 4 -OutPut gene_density
```

This produces the following gene density plot:

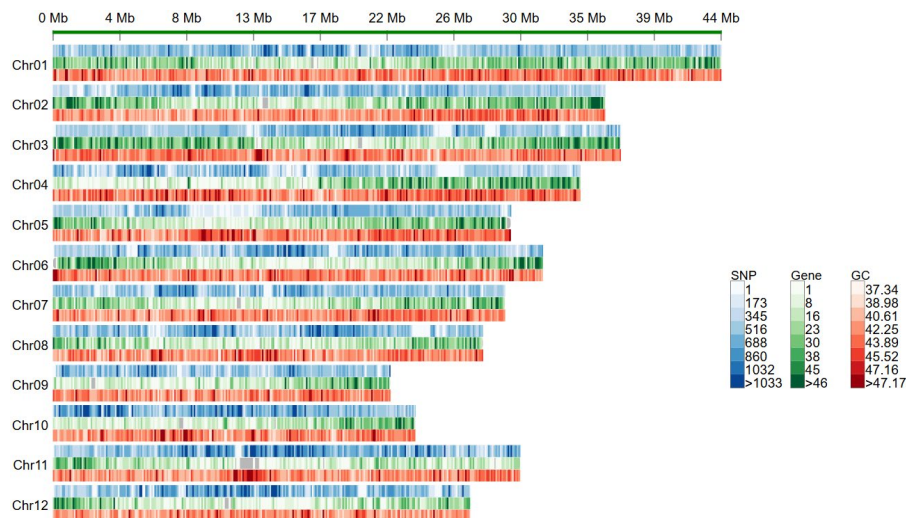


Multiple tracks can be combined, with a different color palette for each track, as shown below.

```
SetParaFor=track1
show_columns=File1:4                                ##确定绘图中需要显示的列, the column where the file is located
colormap_brewer_name=Blues                          ##使用 Tune Rcolorbrewer's Blues palette

SetParaFor=track2
show_columns=File2:4                                ##确定绘图中需要显示的列, the column where the file is located
colormap_brewer_name=Greens                          ##使用 Tune Rcolorbrewer's Greens palette

SetParaFor=track3
show_columns=File3:4                                ##确定绘图中需要显示的列, the column where the file is located
colormap_brewer_name=Reds                          ##使用 Tune Rcolorbrewer's Reds palette
```



In the file format shown below, an additional # adds a title to the legend above.

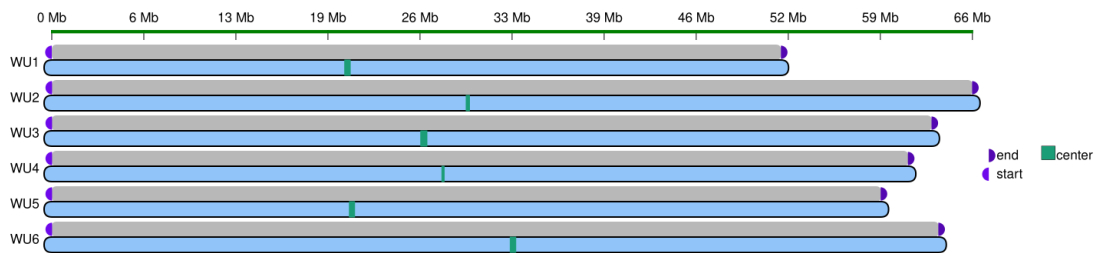
```
#Chr Start End #GC
Chr01 1 100000 44.39
```

4.2 T2T genome mapping

The `track_shift_y` parameter, available in versions after 1.39, was added in response to user requests.

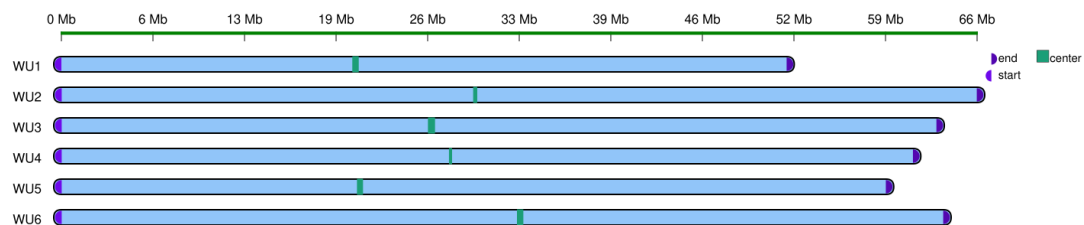
It allows multiple plot types to appear in the same track.

The example below has two tracks: the upper shape track uses two semicircles to mark telomeres, and the lower highlights track marks the centromere.

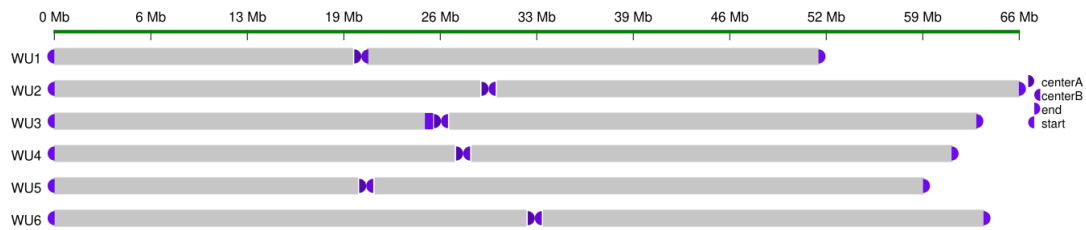


To overlay the two tracks, make the background of track1 transparent and set `track_shift_y=20`. This moves track1 down by 20 units along the Y axis to overlap track2. Add the following settings to track1:

```
bg_stroke_width=0 ##不画此层的, 取消背景色
background_color="#FFFFFF" ##此层(track)背景为白色, 即不画
track_shift_y=20 ##track 下移 和 track2 叠在一起
```



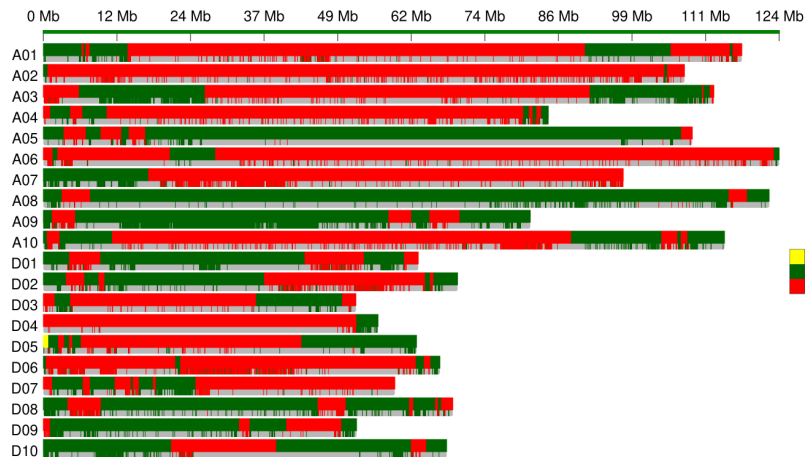
See `in2.conf` and `in3.conf` for detailed settings. The result is shown below.



4.3 Individual BinMap with Makers

Input file1 format: [#Chr Stat EndFlag], where Flag is: P1/P2/H

Input file2 format: [#chr site site Flag] , where Flag is: P1/P2



See the configuration in example4.

```
SetParaFor= global
File1= sample.bin2.gz ##File1 个是必须输入参数,
track_num = 2 ##多少层, how many circles is there for circos?

SetParaFor=track1 ##下面开始处理第 1 layer Parameter processing Swap colors
colormap_low_color="#FFFF00"
colormap_mid_color="#006400" ##此层(ValueX) The middle value Value is the color scheme

SetParaFor=track2 ##下面开始处理第 2 层 参数处理
File2= ./diff.spe.maker.gz ##可以输入别的文件 file1 兼容 gz
show_columns = File2: 4 ##把 file2 的第 4 columns are drawn in highlighter format
track_height=10 ##此层(track)宽度变小点
colormap_low_color="#006400"
```

The plot mainly consists of two heatmap tracks, with the lower track set to `track_height=10`. It shows which parent contributed each region of the offspring genome.

4.4 population BinMap for Multi-region, multi-sample

The previous example shows parental origin for one sample. Breeding populations, such as RIL populations, contain many individuals. Display them together using one heatmap track per sample, zero spacing between tracks, and a horizontal chromosome layout.

- Use one track per sample (`track_num=79`) and set the spacing between tracks to zero (`padding_ratio=0`).
- Order samples by subgroup. Insert several arbitrary samples between subgroups as separators.
- Arrange chromosomes horizontally (`chr_orientation=horizontal`) and reduce the gap between them (`chr_spacing_ratio=0.1`).
- The last column determines the color. Use custom colors or the defaults. See example9 for the

configuration file and example.

```
SetParaFor= global
File1= ./data/S01.bin.gz
track_num = 79 ## 79 samples, each sample has one track, i.e., 79 layers
chr_spacing_ratio=0.1 ##不同染色体 chr 之间的间隔比例
colormap_conf = cor.info ##通过在主配置文件 input Customize the correspondence between color and value
chr_orientation= horizontal ##chr 是按纵 horizontal 横排列

#canvas_height_ratio= 0.7

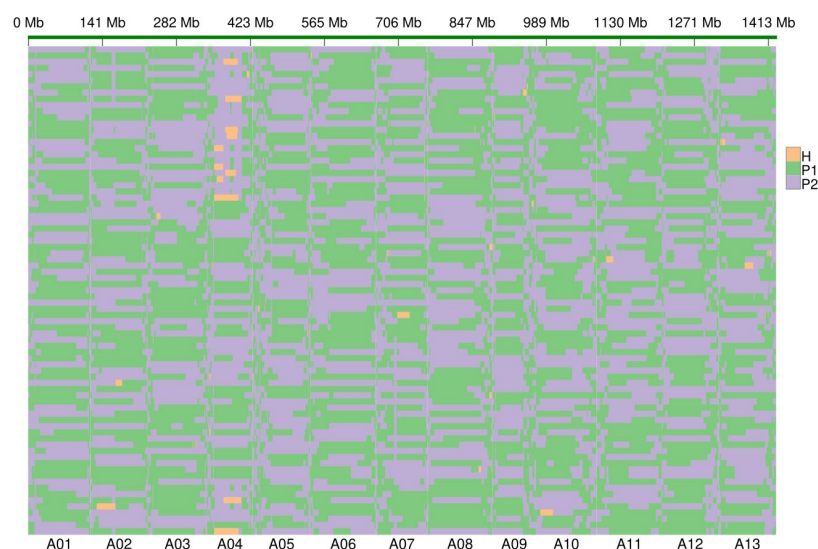
SetParaFor= trackALL ##下面为 trackALL,即先为所有层设置的默认值
plot_type= heatmap
track_height=10 ##79*10 The canvas depth is approximately 790+
colormap_legend_show=0 ##不显示进度条
chr_label_size_ratio=1.05 ##chrName 放大 1.05 倍
show_columns= File1:4

#colormap_legend_shift_y=-550
#colormap_legend_shift_x=20
colormap_legend_size=1.25

SetParaFor=track2
File2=./data/S02.bin.gz
show_columns= File2:4
SetParaFor=track3
File3=./data/S03.bin.gz
show_columns= File3:4

##### Layer Settings for #在这省略 5-77 #####
```

See [example04_RILBinMap](#) for the data and scripts.



4.5 Regional Haplotypes for Multiple Samples

In population resequencing analyses, a region may be selected for closer examination. A regional haplotype plot shows the haplotype distribution across all samples, ordered by subgroup. The script `./bin/script/GenotypeShow.pl` reads a VCF file, converts it to the required format, and calls `RectChr` to generate the plot. Usage is as follows:

```
perl GenotypeShow.pl
Version 1.0 hewm2008 2021-06-28;
Usage: GenotypeShow.pl <InPut><chr:start:end><Out>
Usage: GenotypeShow.pl <InPut><chr:start:end><Sub_or_OrderSample><Out>
Supply in.vcf and the target genomic interval.
```

The optional `Sub_or_OrderSample` argument specifies a one-column file of sample names. Use it to reorder samples or select a subgroup for plotting.

```
perl ../bin/script/GenotypeShow.pl in.vcf.gz Chr2:22759380:22775500 sample.order.list OutRegion
#perl ../bin/script/GenotypeShow.pl in.vcf.gz Chr2:22759380:22775500 OutRegion
```

This produces the following result:



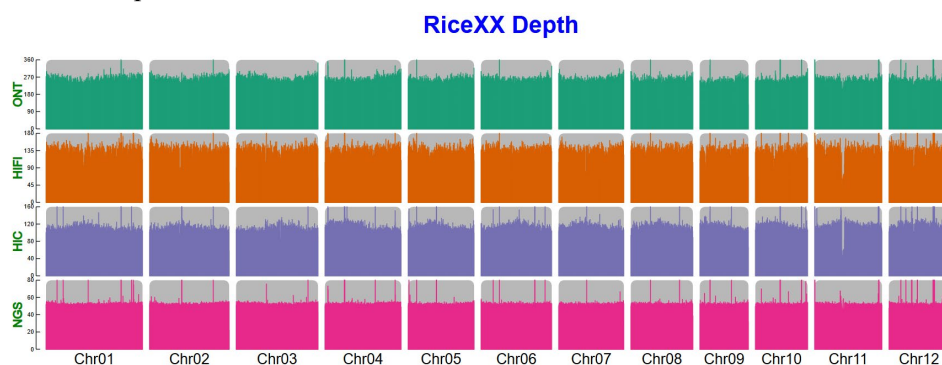
Internally, each sample is treated as a chromosome and displayed vertically, with zero spacing between chromosomes. Thus, the number of displayed chromosomes equals the number of samples. See **example05_RegionHaplotype** and the linked tutorial on regional genotype, gene, and haplotype heatmaps for multiple samples.

4.6 Assembly Coverage Depth Plots

Coverage plots help assess a genome assembly and the depth provided by different datasets. Arrange chromosomes vertically and use one histogram track per dataset. Collapsed copy-number variants, where multiple copies assemble into one, can produce extremely high coverage. Use *cap_max_value* and *cap_min_value* to cap values in a track: values above or below the specified limits are replaced by the corresponding limit. Track1 below illustrates this setting. See **example06_DepthCov** for more details.

```
SetParaFor=track1
show_columns=File1:4
colormap_low_color="#1B9E77"
colormap_mid_color="#1B9E77"
colormap_high_color="#1B9E77"
label= "ONT"
cap_max_value=360
#Ymax=360
#bg_end_arc_division=98
```

This produces the plot below:

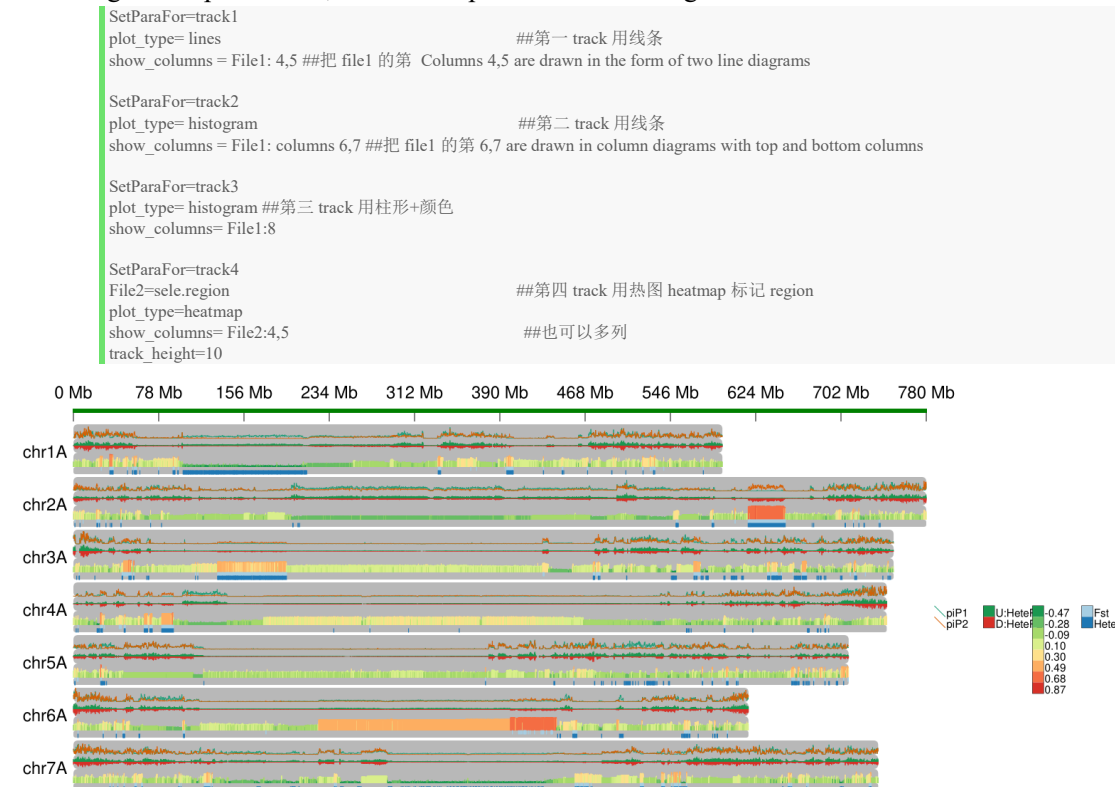


4.7 Combined Plots of Genetic Statistics

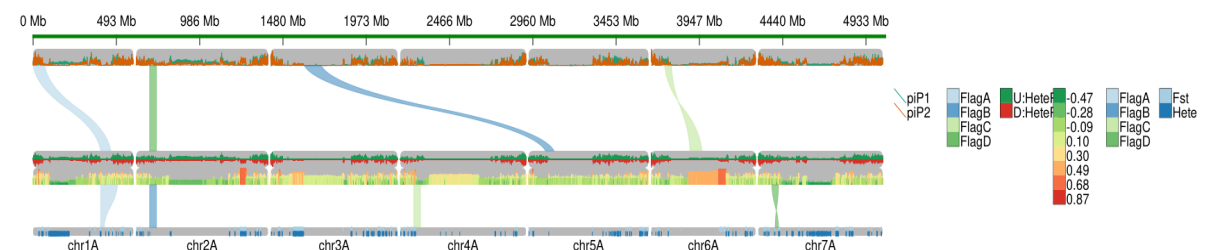
Tracks can be combined freely. The example below uses three tracks; see [example07_Genetics](#). A: Lines represent P_i .

B: Bar height and color represent heterozygosity.

C: Histograms represent F_{st} , and heatmaps mark selected regions.



To combine this plot with LinkS in a horizontal layout, see *in2.conf*. The resulting plot is shown below.



4.8 Enhanced GWAS Manhattan Plots

The scatter/point section introduced a standard GWAS plot. To label genes at association peaks, add two tracks to that configuration: one text track and one PairWiseLink track.

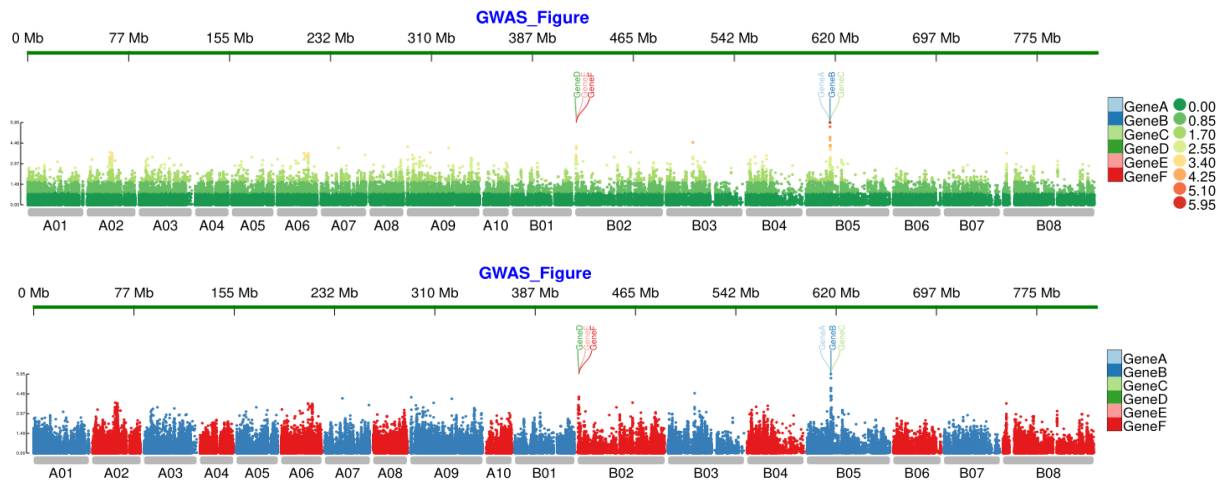
```
SetParaFor=track1                                ##下面开始处理第 1 层 参数处理
plot_type= text                                  ##文件 File2 的第 5 列作图
show_columns= File2:5
background_color="#FFFFFF" ##白色, meaning no background is drawn
track_height = 40 ##放 text 的名字, so the width is a bit longer
#track_text_angle=-90 ##文本旋转 90 度
```

```

track_text_size=1.2                                ##文字字体放大为原来的 1.2
colormap_legend_show=0 ##不显示渐变条图列

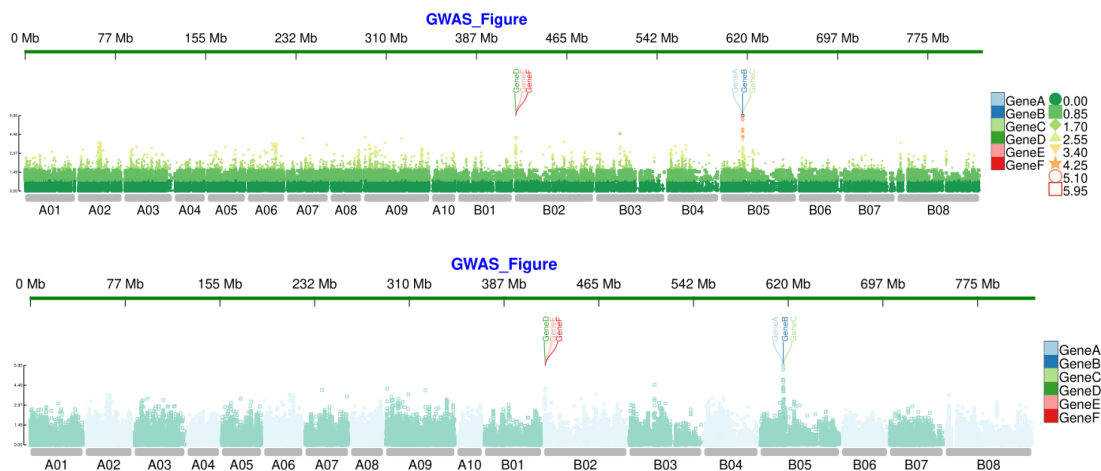
SetParaFor=track2
show_columns=File2:5                                ##把 file2 的第 5 列用 PairWiseLink S 曲线
track_height=30                                     ##这一层宽度调高点
plot_type=PairWiseLink                             ## 贝塞尔曲线
link_direction=UpDown
background_show=0                                  ##同 background_color="#FFFFFF"一样，即没有背景
colormap_legend_size = 0.8 ##渐变条 reduced to the original 0.8

```



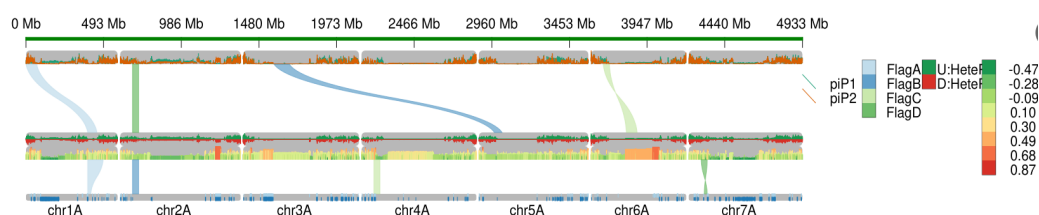
See run3.sh for an example that replaces the point plot with shapes.

For shape parameters, see the section below or open the linked description.



4.9 Regional Zoom with zoom_region

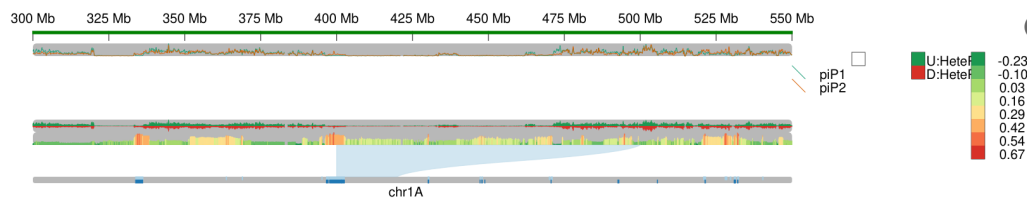
In earlier versions, chr_order under SetParaFor=global could reorder chromosomes or select one or more chromosomes, but could not select a subregion. Add zoom_region (equivalent to chr_zoom_region) under SetParaFor=global and specify the region as ChrA:StartA:EndA.



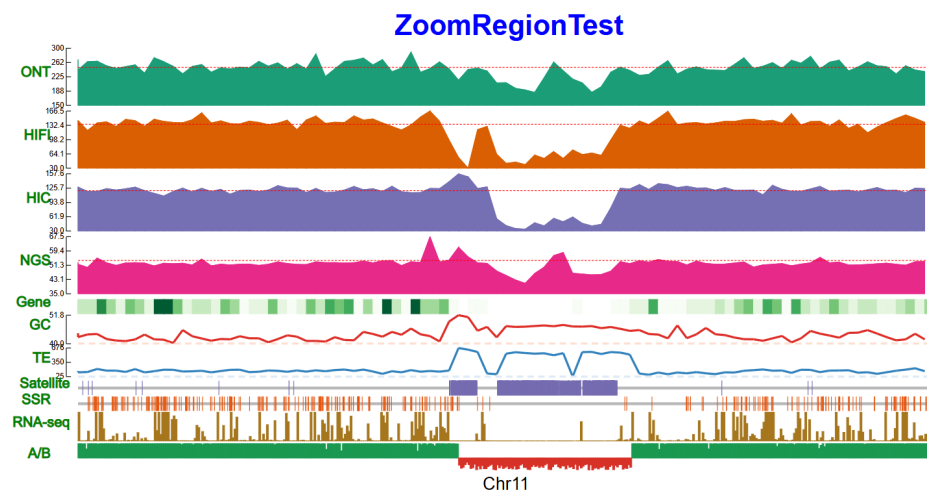
For example, add the following to in2.conf in application example 5.7:

```
zoom_region=chr1A:300000000:550000000 #即可以只放大这个区域的情况。
```

The plot below displays only the 300–550 Mb region of Chr1A, as indicated by the axis.



The **example09_ZoomRegion** directory includes the following publication-style plot, with the centromere in the middle.



上图是在 `zoom_region=Chr11:8000000:17000000` 显示局部情况。

Example10_Peak illustrates gene structures and relationships within a specific region, as shown below.



5. Additional Settings

5.1 Color Settings

A: Earlier versions of RectChr assigned colors using three gradient colors, `colormap_low_color`, `colormap_mid_color`, and `colormap_high_color`, together with the number of color levels,

colormap_nlevels.

```
#colormap_nlevels=3
#colormap_low_color = "#006400" ## The color corresponding to the lowest value
#colormap_mid_color = "#FFFF00" ## The color corresponding to the middle value
#colormap_high_color = "#FF0000" ## The color corresponding to the highest value
```

B: Later versions added color palettes through *colormap_brewer_name* and *colormap_reverse* for each track. The original three-color gradient is still supported.

When *colormap_brewer_name* is set, it overrides the three-color gradient.

The three-color settings, *colormap_low_color*, *colormap_mid_color*, and *colormap_high_color*, take effect only when *colormap_brewer_name* is not specified.

B.1 Built-in palettes include those from RColorBrewer. See its website: [R Color Brewer's palettes](http://colorbrewer2.org/).

RColorBrewer also provides an interactive palette website: [Color Advice for Maps](http://coloradvice.org/).

Palette keywords and their color scales are shown below.

```
"BrBG", "PiYG", "PRGn", "PuOr", "RdBu", "RdGy", "RdYlBu", "RdYlGn", "Spectral"
"Accent", "Dark2", "Paired", "Pastel1", "Pastel2", "Set1", "Set2", "Set3"
"Blues", "BuGn", "BuPu", "GnBu", "Greens", "Greys", "Oranges", "OrRd", "PuBu", "PuBuGn", "PuRd", "Purples", "RdPu", "Reds", "YlGn", "YlGnBu", "YlOrBr", "YlOrRd"
"BrBG", "PiYG", "PRGn", "PuOr", "RdBu", "RdYlBu", "Dark2", "Paired", "Set2", # 默认为 GnYlRd 绿黄红
```



B.2 Before version 1.42, *colormap_brewer_name=xx* accepted only predefined RColorBrewer names such as Set3 and Dark2. You can now place a file containing a color sequence, such as bs5, in the ColorsBrewer directory and set *colormap_brewer_name=bs5* to use a custom palette. The bs5 format is shown below. Color schemes from ggsci are also included in ColorsBrewer.

```
NewParaList.xlsx Scene_Usage
(base) liaojing@node1:/home/liaojing/07.tmp/RectChr-1.42
$ ls ColorsBrewer/
aaas      d3         gsea      locuszoom primer      tron
atlassian flatui     igv        material  rickandmorty tw3
bmj        frontiers jama       nejm      simpsons    uchica
bs5        futurama  jco        npg        startrek    ucscgb
cosmic     get_colors.pl lancet     observable tmp
(base) liaojing@node1:/home/liaojing/07.tmp/RectChr-1.42
$ head ColorsBrewer/bs5
#f8f9fa
#e9ecef
#dee2e6
#ced4da
#adb5bd
#6c757d
```

C: To map values to colors, the program detects numeric or ASCII sorting (which can also be set manually), divides the range into levels, and assigns colors. Range limits and related parameters can be specified. Alternatively, add `colormap_conf=col.fille` to the global settings to supply a custom value-to-color mapping. The format is shown below.

```
Value1=red
Value2="#f8f9fa"
...
```

5.2 Legend Layout

Versions after 1.41 include `colormap_legend_layout`, with a default of 1 and 14 options numbered 0–13. The canvas adjusts automatically to the number of legends. The layouts are shown below.

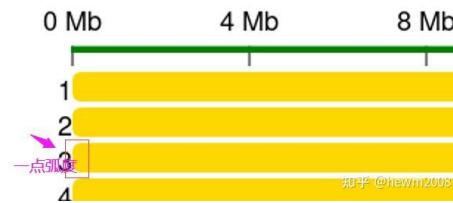
Value	Illustration	Value	Illustration	Value	Illustration
0		1 Default		2	
3		4		5	
6		7		8	
9		10		11	
12		13		Others	Same as 0, do not draw legend

`colormap_gradient_gap` controls the spacing between successive legends. `colormap_legend_gap` controls the vertical spacing between two Flag entries within the same legend.

Other related legend parameters can also be used: `colormap_legend_show/colormap_legend_shift_x/colormap_legend_shift_y/colormap_legend_size`

5.3 Chromosome End Curvature

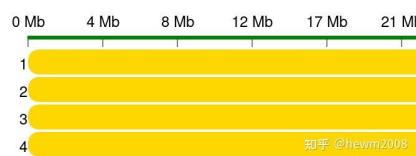
By default, RectChr subdivides the chromosome ends to produce a slight curve, approximately as shown below.



Set *bg_end_arc*=0 in the relevant track to remove this curvature:



To create more strongly curved ends, as shown below: The default divides the end into three segments. Changing the division to two produces a semicircle. Set the hidden parameter *bg_end_arc_division*=2. The minimum value is 2; very large values approach the flat-ended result of *bg_end_arc*=0.



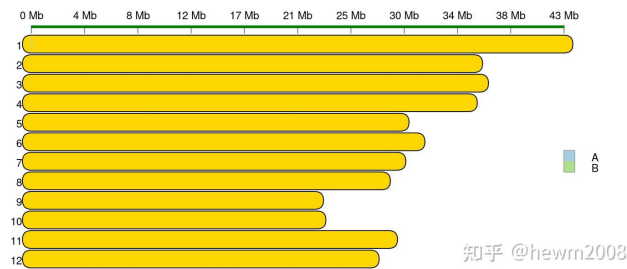
However, a vertical line at 10 bp may appear to cut through the semicircle, as illustrated below.



In the desired example, the semicircular ends lie outside the genomic coordinate system. Version 1.30 added the hidden parameter *bg_end_offsset*=1 to position them outside the coordinate range and avoid this overlap. This setting works only when chromosomes are arranged vertically.



Use *bg_end_arc_division*=2 together with *bg_end_offsset*=1. Then set *bg_stroke_width*=2 for the background outline width and *bg_stroke_color*=black for its color to obtain the following result



5.4 X Axis Settings

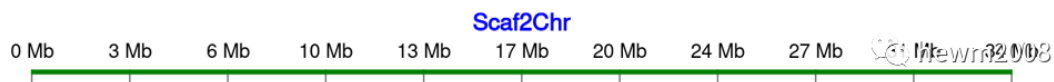
X-axis related parameters:

<code>#axis_tick_show=1</code>	<code>## bool 0-1</code>	Whether the X-axis is displayed
<code>#yaxis_tick_show=0</code>	<code>## bool 0-1</code>	Y-axis display
<code>#axis_tick_unit</code>	<code>## Text, for customization, such as kb, bp, cm, mb</code>	Coordinate axis scale units, automatic determination
<code>#axis_text_angle=0</code>	<code>## Integer (-360,360)</code>	Coordinate axis text angle
<code>#axis_tick_num=10</code>	<code>## Integer</code>	Coordinate axis scale count
<code>#axis_tick_interval</code>	<code>## Integer</code>	Coordinate axis scale intervals, such as intervals every 10 meters; if started,
<code>#axis_tick_num_fail</code>	<code>## Integer</code>	Coordinate axis scale accuracy, automatic judgment
<code>#axis_tick_precision</code>	<code>## Integer</code>	Coordinate axis scale accuracy, automatic judgment
<code>#xaxis_shift_y</code>	<code>## Values</code>	X-axis vertical offset

To hide the axis, set the global tick count, which defaults to 10, to `axis_tick_num=-1`, or set `axis_tick_show=0`.

If the longest chromosome is 111 Mb, 10 equal divisions produce ticks at 0, 11, 22, and so on. To use a custom interval of 20 Mb instead, add `axis_tick_interval=20000000` to the global settings. The hidden `axis_text_angle` parameter can also rotate tick labels.

The default plot below divides the longest chromosome, 34 Mb, into 10 equal intervals.



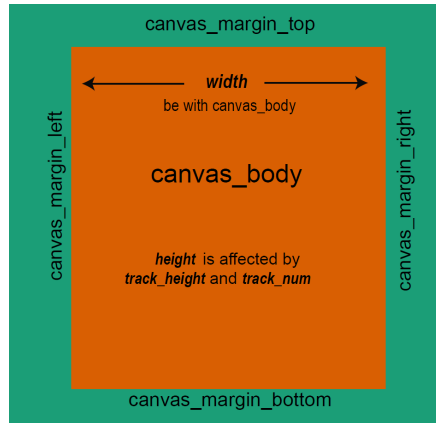
Adding `axis_tick_interval=5000000` to the global settings gives:



5.5 Canvas Settings

Canvas parameters:

<code>#canvas_body = 1200</code>	<code>## Main canvas size, default value is 1200</code>
<code>#canvas_margin_top = 55</code>	<code>## Margin on the canvas, default is 55</code>
<code>#canvas_margin_bottom = 55</code>	<code>## Margin at the bottom of the canvas, default is 25</code>
<code>#canvas_margin_left = 100</code>	<code>## Left margin of the canvas, default is 100</code>
<code>#canvas_margin_right = 120</code>	<code>## Canvas's right margin, default is 120</code>
<code>#canvas_width_ratio=1.0</code>	<code>## 画布 width 的缩放比例</code>
<code>#canvas_height_ratio=1.0</code>	<code>## 画布 height 的缩放比例</code>



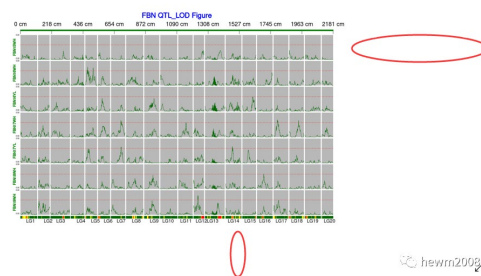
Long chromosome names on the left may be clipped, as shown below.



Increase `canvas_margin_left` to provide enough space.

`canvas_height_ratio` = 1.0 and # `canvas_width_ratio` = 1.0

These settings can reduce blank space below and to the right of the plot.



In multitrack plots, `colormap_legend_show=0` hides each track's legend. Earlier versions could leave substantial blank space on the right; this no longer occurs after version 1.41. In the eight track example below, all color scales are hidden, leaving blank space on both sides. Add the following two lines to the global settings.

```
canvas height ratio =0.8
canvas_width_ratio =0.7
```

This scales the height to 0.8 and the width to 0.7 of the original canvas.

If part of a legend is clipped, increase the corresponding scale factor above 1.

6. Download and Installation

6.1 Download

Updates are available at <https://github.com/hewm2008/RectChr> . Please consider starring the repository.

6.2 Prerequisites

RectChr runs on Linux, Unix, and macOS. Prepare the following dependencies before installation:

- 1) Perl and its SVG.pm module (<https://metacpan.org/release/SVG>). SVG.pm can be installed separately, although it is already bundled with the package.
- 2) The convert system command, used to convert SVG to PNG. This is optional but useful.

6.3 Installation

On Linux or macOS, set executable permissions with *chmod 755* and run the program directly:

```
git clone https://github.com/hewm2008/RectChr
cd RectChr ; chmod 755 -R bin/*
./bin/RectChr -h #
```

7. Usage and parameters

7.1 RectChr Parameters

RectChr requires a configuration input and an output path, as follows.

```
[heweiming@cngb-ologin-25 bin]$ ./bin/RectChr
Version:1.42          hewm2008@gmail.com
```

Options

-InConf	<s>	:	Input Configuration File
-OutPut	<s>	:	OutPut svg file result
-help		:	Show more help with more parameter

- InConf Input file: The configured file
- OutPut Output file, out.svg
- help See configuration parameters for details

The program performs basic checks on the input configuration. It reports errors if the configuration or input file format is invalid, including inconsistent numbers of columns. Review the messages and correct the corresponding settings.

7.1.1 Main configuration parameters

SetParaFor selects the scope of the following parameters: global for the whole plot, ALL for all tracks, or trackX for track X.

The main configuration parameters are explained below.

Serial number	Example	Explanation
Parameter 1	FileN = /path/in.data	(N=1,2,3...) Enter the path of the Nth file
Parameter 2	Plot_type = heatmap	[关键词中的一种: heatmap/ highlights/LinkS/ pairwiselink/ histogram/hist/ scatter/ point/shape/text/line/lines/heatmapAnimated 和 histAnimated, 即共有 13 种画法] If it is not a key word, it is not drawn, only a background strip (See the previous sections for detailed drawing methods 3.Example of drawing methods)
Parameter 3	show_columns = File1:8 File2:5,6	The Xth column of a file is used as a statistic

The screenshots below show other important parameters.

Scope	Parameter	Importance	Type / Value	Summary	Description
track	colormap_low_color	**	Text: color	Gradient start color	Set the starting color of the plot's color gradient.
track	track_geom_shape	**	Integer (0-13)	Shape type	Set the shape used in the plot, such as a circle or square.
track	link_direction	**	Text: keyword	Link direction	Set the up/down direction of plot elements.
track	link_linestyle	**	Text: keyword	Line style	Set the line style, such as solid or dashed.
track	chr_scale_ratio	***	Floating-point ratio	Chromosome length scale	Scale chromosome length relative to a reference.
global	title	***	Text	Plot title	Set the plot title, such as a label or heading.
track	log_p	***	Boolean (0-1)	Log-transform P values	Process P values displayed on a logarithmic scale.
track	cap_max_value	***	Numeric	Upper cap	Set an upper cap to limit the displayed data range.
track	cap_min_value	***	Numeric	Lower cap	Set a lower cap to limit the displayed data range.
track	ymin	***	Numeric	Minimum value	Set the minimum value for the plotted data.
track	ymax	***	Numeric	Maximum value	Set the maximum value for the plotted data.
track	padding_ratio	***	Floating-point ratio	Track spacing ratio	Control the gap between tracks as a ratio.
track	track_height	***	Integer	Track height	Set the height of the track.
track	label	***	Text	Track label	Set the name of the track.
global	colormap_conf	***	Text: file path	Color configuration	Configure plot colors, including line and fill colors.
global	chr_order	****	Text: chr01,chr02,....	Chromosome order	Set the chromosome display order in the figure.
track	chr_order	****	Text: chr01,chr02,....	Chromosome order	Set the chromosome display order at the track level.
global	chr_zoom_region	****	Text: file path	Chromosome region zoom	Zoom into a specified chromosome region to inspect details.
track	colormap_brewer_name	****	Text: keyword	Brewer color palette	Select a predefined palette for the plot.
track	colormap_nlevels	****	Integer	Number of color levels	Set the number of color levels in the legend.
global	chr_orientation	*****	Text: horizontal or vertical	Chromosome orientation	Set whether chromosomes are arranged horizontally or vertically.
track	plot_type	*****	Text: keyword	Plot type	Set the plot type used to display the selected statistic.
track	show_columns	*****	Text: keyword, e.g. File1:4	Columns to display	Specify which input columns to display.
global	track_num	*****	Integer	Number of tracks	Set the number of tracks in the plot.

7.1. 2 Other configuration parameters

All relevant parameters can be customized. Many have defaults, but can be specified explicitly.

For the complete parameter list, see the Excel file.

chr_label_shift_x	chr_label_shift_x	Integer	Chromosome label horizontal offset	Adjust chromosome label position horizontally.	
chr_label_shift_y	chr_label_shift_y	Integer	Chromosome label vertical offset	Adjust chromosome label position vertically.	
chr_order_reverse	chr_order_reverse	Boolean: reverse	Reverse chromosome order	Reverse the chromosome display order.	
axis_tick_unit	axis_tick_unit	Text: custom unit, e.g. kb or bp	Axis unit	Set the axis unit and the numerical interval.	
axis_text_angle	axis_text_angle	Integer (-360,360)	Axis text angle	Set tick-label rotation to a specified angle.	
axis_tick_show	axis_tick_show	Boolean 0-1			
axis_tick_num	axis_tick_num	Integer	Number of axis ticks	Set the number of axis ticks.	
axis_tick_interval	axis_tick_interval	Integer	Tick interval	Set the numerical interval between axis ticks.	
axis_shift_y	axis_shift_y	Numeric	Vertical axis offset	Adjust the vertical axis position.	
axis_tick_precision	axis_tick_precision	Integer	Tick-label precision	Set the number of decimal places in tick labels.	
yaxis_tick_show	yaxis_tick_show	Boolean 0-1	Show Y axis	Control whether the Y axis is displayed.	
title_color	title_color	Text: color	Title text color	Set the main title color.	
title_size	title_size	Floating-point ratio, may exceed 1	Title size scale	Scale the title text relative to its original size.	
title_shift_x	title_shift_x	Integer	Title horizontal offset	Adjust the horizontal title position.	
title_shift_y	title_shift_y	Integer	Title vertical offset	Adjust the vertical title position.	
Internal parameter, not exposed	Internal parameter, not exposed	Not fixed: internal	Numeric or text	May be used to determine whether an element or dataset is numeric or textual	
as_flag	as_flag	Text: keyword	Label	Can serve as a marker to control a plot feature or state.	
upper_outlier_ratio	upper_outlier_ratio	Floating-point ratio (0-1)	Upper outlier ratio	Handle values above a specified quantile in the data.	
lower_outlier_ratio	lower_outlier_ratio	Floating-point ratio (0-1)	Lower outlier ratio	Handle values below a specified quantile in the data.	
background_color	background_color	Text: color	Background color	Set the plot background color.	
background_show	background_show	Boolean 0-1	Show background	Control whether the background is displayed.	
bg_end_arc	bg_end_arc	Boolean 0-1	Rounded chromosome ends	Control the shape of the chromosome ends.	
bg_end_offset	bg_end_offset	Boolean 0-1	End offset	Control the offset at chromosome ends.	
bg_end_arc_division	bg_end_arc_division	Integer	End subdivisions	Set the number of subdivisions at chromosome ends.	
track_bg_height_ratio	track_bg_height_ratio	Integer	Background height ratio	Set background height relative to chromosome height.	
track_label_color	track_label_color	Text: color	Track label color	Set the track label text color.	
track_label_size	track_label_size	Text: color	Track label size scale	Scale the track label text size.	
track_label_angle	track_label_angle	Numeric (-360,360)	Track label angle	Set the track label rotation.	
track_label_shift_x	track_label_shift_x	Integer	Track label horizontal offset	Adjust the horizontal track label position.	
track_label_shift_y	track_label_shift_y	Integer	Track label vertical offset	Adjust the vertical track label position.	
track_colormap_reverse	track_colormap_reverse	Colors below can also be specified as RGB values.	Reverse color gradient	Reverse the color-gradient direction.	
track_colormap_mid_color	track_colormap_mid_color	Text: color	Gradient middle color	Set the middle color of the plot gradient.	
track_colormap_1egh_slow	track_colormap_high_color	Text: color	Gradient end color	Set the ending color of the plot gradient.	

SetParaFor=X selects the scope for subsequent settings. global applies to the whole plot.

ALL sets defaults for all tracks. A track inherits these values unless it overrides them. Track X corresponds conceptually to ring X in Circos.

The example below explains settings in comments following ##.

Additional parameters that are less frequently used are hidden, such as the vertical offset of the X axis.

For example, use `xaxis_shift_y=0` under global. If more parameters are needed, please provide feedback; the program was designed to allow parameters to be added or removed.

```
##### Global Parameters #####

SetParaFor = global ## Set the current paragraph scope to global scope, applicable to the entire graphical configuration

File1 = ./InPut.file ## Main data file path,
#File2 = ## Optional second data file path for multi-source data overlay FileX = ./InPut.fileX

#track_num = ## Specify the number of drawing layers (number of tracks), which by default automatically infers based on the number of
columns in File1 (e.g., Value1 to ValueN)
#chr_spacing_ratio = 0.2 ## Proportion of intervals between different chromosomes, calculated based on track height (track_height *
chr_spacing_ratio)
#title = "main Figure"          ## 图形标题及其样式设置, 支持 title_color、title_size、title_shift_x、title_shift_y 等参数
#colormap_conf = col.file ## Custom color mapping file path, used to specify the mapping relationship between value and color (e.g., P1 =
"#FE0808")
#chr_orientation = horizontal ## Chromosome arrangement direction, optional values can be horizontal or vertical
#bg_end_arc = 1 ## Less commonly used parameters, used to control special effects such as background arcs

##### global. Chromosomal Parameter Configuration #####
#zoom_region = ## 缩放特定区域, 格式为 chr:start:end(例如 chr2:1000:5000)
#chr_order = ## Specify chromosome order or filter the displayed chromosome list (if not specified, automatically sort by name)
#chr_spacing_ratio = 0.2 ## The spacing ratio between different chromosomes
#chr_label_rotation = 0 ## Chromosome label text rotation angle

##### global. Canvas and Image Parameter Configuration #####

#canvas_body = 1200 ##主画布大小, default value is 1200
```

```

#canvas_margin_top = 55 ##画布上边距, default is 55
#canvas_margin_bottom = 25 ##画布下边距, default is 25
#canvas_margin_left = 100 ##画布左边距, default is 100
#canvas_margin_right = 120 ##画布右边距, default is 120
#canvas_angle = 0 ##整个图形旋转角度, unit: degrees
#canvas_height_ratio=1.0 ##收缩放画布的整体高度
#canvas_width_ratio=1.0      ##收缩放画布的整体宽度

##### ALL Default configuration parameters for each layer. If the layers are not configured, the parameters here will be
used. #####
SetParaFor = trackALL ## Set the default parameters for all tracks; trackX not configured separately in the future will carry over these settings
plot_type = heatmap ## Supported drawing types include: heatmap, line and a total of 13 types, Check here for details
#show_columns = ## Specify the columns to display, such as File1:4 or File2:4,5
#colormap_brewer_name =      ## 使用预设 Rcolorbrewer 调色板, 例如 GnYlRd (数值) 或 Paired (文本)
#colormap_reverse = 0 ## Whether to invert the color gradient bar (0=normal, 1=invert)
#colormap_low_color = "#006400" ## The color corresponding to the lowest value
#colormap_mid_color = "#FFFF00" ## The color corresponding to the middle value
#colormap_high_color = "#FF0000" ## The color corresponding to the highest value
#background_color = "#B8B8B8" ## Background color
#upper_outlier_ratio = 0.95 ## Define the top outlier threshold; values above this ratio use the highest color
#lower_outlier_ratio = 0 ## Define the bottom outlier threshold; values below this ratio use the lowest color
#Ymax = ## Manually set the maximum value of the current layer to override the automatically calculated result
#Ymin = ## Manually set the minimum value of the current layer to override the automatically calculated result
#cap_max_value = ## Limit the maximum value
#cap_min_value = ## Limit the lower limit of the minimum value
#colormap_nlevels = 8 ## Number of gradient color levels
#track_height = 20 ## Current layer height (i.e., orbital height)
#track_bg_height_ratio = 1 ## Proportion of background height to track height (0-1)
#log_p = 0 ## Whether to perform 0-log10 conversion on the value (0=No, 1=Yes)
#padding_ratio = 0 ## The ratio of vertical spacing between adjacent tracks within the same chromosome
#colormap_legend_sizeratio = ##渐变条尺寸比例收缩放
#yaxis_tick_show = 0 ## Whether to display Y-axis scale labels (0=hide, 1=show)
#colormap_legend_show = 1 ## Whether to display color gradient bars (0=hide, 1=show)
#colormap_legend_shift_x = 0 ## Horizontal offset of the gradient bar
#colormap_legend_shift_y = 0 ## Vertical offset of the gradient bar
#chr_label_shift_x = 0 ## Lateral offset of chromosome tags
#chr_label_shift_y = 0 ## Longitudinal offset of chromosome tags
#chr_label_size_ratio = 1.0 ## Chromosome tag font size ratio (relative to default value)
#track_shift_x = 0 ## Move the X value coordinates of the track layer
#track_shift_y = 0 ## Move the Y coordinates of the track layer

##### trackALL. Other uncommon parameters #####

#text-font-size = ## Set the font size for the text
#track_text_size = 1.0 ## Text font size ratio (relative to default value)
#... ##### More parameters

##### Parameters of each trackX layer, inheriting all parameters from trackALL #####

#SetParaFor = track2 ## Start processing the second-layer parameters
#File2 = ## You can specify other input files as data sources
#plot_type = hist ## 绘图类型: histogram (直方图)
#show_columns = File2: 5 ## Displays the fifth column of data in File2
#label = "Name"      ## 当前层的标签及其样式设置 (label_size label_color ... )
#SetParaFor = track3
#plot_type = lines ## Drawing type: line chart (lines)
#show_columns = File1: 5,6 ## displays data in columns 5 and 6 of File1

```

7.2 Input Files

At least one data file is required. Multiple input files and gz-compressed files are supported.

```

File1=path1/in1.file
File2=path2/in2.file.gz
...
FileN=pathx/inX.file

```

Chromosome lengths are inferred from the first three columns of these files.

7.2.1. Required Data File

Input must be a text file with the same number of columns on every data row. The first three columns must follow the required format.

The plot input format is:

[Chr name Coordinates 1 and 2 F lag1 Flag2 ...]

If the first line is a header, it must begin with #. Sorting is not mandatory, but is recommended before plotting: sort -k 1,1 -k 2,2n.

4. 2. 2 Optional Color Configuration

An optional file can define custom colors. The format is simple and uses hexadecimal HTML color codes.

```
A="#F8F8F3"
B="#A3FF88"
C="#FFFFFF"
...
...
```

Value-to-color mapping

Without a custom color file, colors are assigned automatically according to the selected ColorBrewer palette, the number of distinct values, the number of color levels, the frequency of each value, and the minimum and maximum values. See the linked explanation for details. chr Start End Flag(Value)

If all Flag entries are numeric, such as 1, 2, or 1e10, the program treats them as numbers (as=numeric). If any entry contains letters or other nonnumeric content, the entries are treated as strings (as=factor).

The default mapping rules are illustrated below. For example:

Assuming 9 equal colors, there are 100 Flags, with the following frequencies for Flag statistics

01: 10
02: 20
03: 20
04: 20
05: 20
06: 10
80: 4
84: 2
99: 4

Then:

If all entries are numeric, the maximum is 99 (or a value supplied through YMax) and the minimum is 01 (or a value supplied through YMin).

When *upper_outlier_ratio*=1:

01–06 map to color 1; 80 and 84 map to color 8; 99 maps to color 9. When *upper_outlier_ratio*=0.95:

The six entries with values 84 or 99 exceed 5% of all entries, so both values map to the highest color, color 9. Values 01–05 map to color 1, and 80 maps to color 8.

If all entries are treated as strings:

Nonnumeric Flag entries are treated as strings.

Numeric values can also be treated as strings using the hidden parameter *asFlag=*factor. When *upper_outlier_ratio=1*:

There are nine distinct values and nine color levels, so each value maps to one color in ASCII order: 01 to color 1, 02 to color 2, and so on through 99 to color 9.

When *upper_outlier_ratio=0.95*:

In ASCII order, the six entries with values 84 or 99 exceed 5%, so both map to color 9. The remaining seven values have eight colors available; because there are more colors than categories, seven evenly spaced colors are used, one per value. Eight colors are used in total.

For additional background_color settings, such as removing the fill while retaining the outline, see the linked explanation. So

7.3 Output Files

Output file	Explanation
out.svg	Exported SVG format image
out.png	PNG format diagram for output

The sample diagram is shown in the above application scenario

8. Examples

Examples are available for the application scenarios described above. See the following for data formats and configurations:

```
./bin/RectChr -InConf in.conf -OutPut out.svg
```

Eight application examples are introduced here, with configuration files and input formats. They can be run in a few minutes.

See **Basic_Tutorials/example*** and **Scene_Usage/example*** in the software package.

Data and configurations are provided in the example* directories. The following links also collect configurations and illustrations shared by other users online.

[RectChr General Introduction](#)

[RectChr Comparison Between Two Groups Selected Signal + Selected Region](#)

[RectChr: A population genetic polymorphism](#)

[Several RectChr methods are selected in the region](#)

[RectChr Comparison of Two Groups Selectable Signal + Selected Area Horizontal CHR](#)

[RectChr 之 群体 sweep +QTLs+IBD 作图](#)

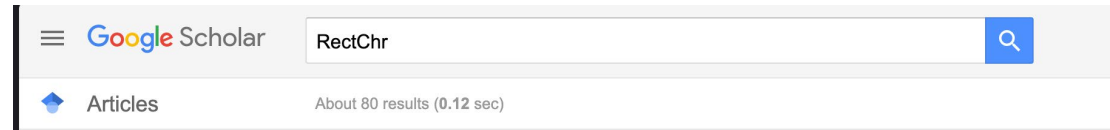
[RectChr's two-genome collinearity analysis](#)

[RectChr's multi-genome \(3 or more\) collinearity analysis](#)

[RectChr's multi-sample genotype/gene/haplotype heat map for specific regions](#)

[RectChr's multi-sample, multi-area binmap heat map](#)
[of RectChr Dynamic heat charts and dynamic bar charts](#)
[of RectChr The sophisticated GWAS Manhattan map](#)

Additional examples are updated on the website: See <https://github.com/hewm2008/RectChr>
Published papers also provide examples of applications.



9. Advantages

- 1 Fast plotting with low memory use and support for batch processing.
- 2 Extensive customizations of chromosome tracks, colors, heights, and layout for different applications.
- 3 A broad range of applications: as with Circos, users can combine their own datasets in a plot.
- 4 No installation required; straightforward to use.
- 5 There are at least 80 citations



10. Frequently Asked Questions

10.1 Changes to RectChr Parameter Names

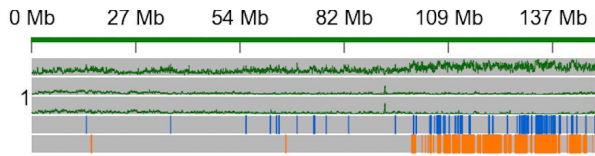
Before version 1.39, features and parameters were added incrementally in response to user requests, resulting in inconsistent naming. Parameter names were standardized from version 1.39 onward. A table maps old names to new names, and ParaOld2New.pl converts existing configuration files to the new format.

```
Perl ParaOld2New.pl in_old.conf out_new.conf
```

10.2 Hiding Axes or Setting Custom Tick Intervals

To hide the axis, set the global tick count, which defaults to 10, to `axis_tick_num=-1`, or set `axis_tick_show=0`. See the X Axis Settings section above.

10.3 Using a Single Color for Text or Highlights



Set `colormap_nlevels=3` in the relevant track and set all three gradient colors to the same color, as shown below

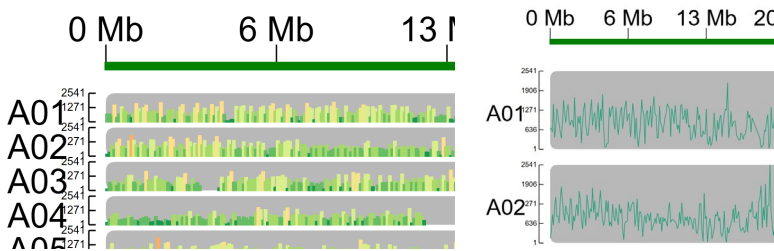
```
#colormap_nlevels=3
#colormap_low_color = "#006400" ## The color corresponding to the lowest value
#colormap_mid_color = "#FFFF00" ## The color corresponding to the middle value
#colormap_high_color = "#FF0000" ## The color corresponding to the highest value
```

10.4 Displaying Axis Labels for Lines Points and Histograms

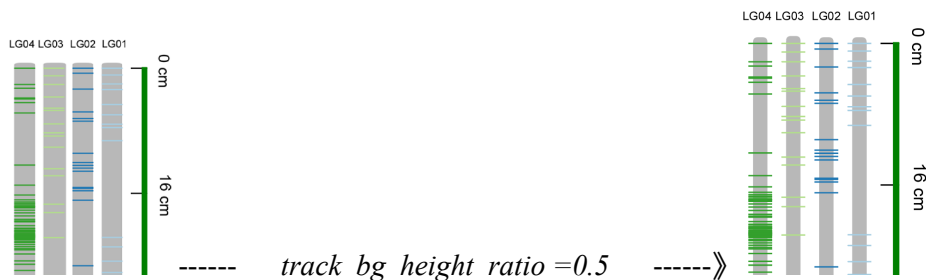
Can numeric values be added to the left of the line plot shown above? The following parameter was added in versions after 1.24:

```
yaxis_tick_show=0 ## Whether to display the start and end values of the Y axis for all layers, the default value is this: 0 Not displayed
```

Set `yaxis_tick_show=1` in the relevant track. See the parameter list for additional `yaxis_tick` properties.



10.5 Extending Genetic Map Markers Beyond the Background



By default, the background width `bg_height` is the same as `track_height`. Reduce the following setting: This shrinks `bg_height` toward the center.

```
#track_bg_height_ratio = 1 ## Proportion of background height to track_height (0-1)
```

10.6 Preventing Overlapping Text in Densely Labeled Regions

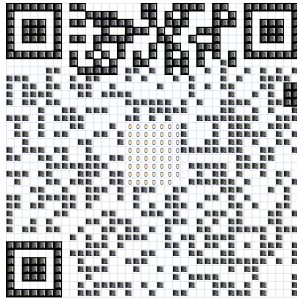
Versions after 1.39 include `track_text_overlap`, which defaults to 5. Labels are kept from overlapping, with at most five labels displayed in a region; additional labels are omitted, as shown below.

10.9 Contact and Donations

Donations are entirely optional. Thank you for your support.

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WeChat tipping QQ group joining: 125293663 *WeChat public account*



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